PART IV

Rings and Fields

Section 18 Rings and Fields
Section 19 Integral Domains
Section 20 Fermat's and Euler's Theorems
Section 21 The Field of Quotients of an Integral Domain
Section 22 Rings of Polynomials
Section 23 Factorization of Polynomials over a Field
Section 24 †Noncommutative Examples
Section 25 †Ordered Rings and Fields

SECTION 18 RINGS AND FIELDS

All our work thus far has been concerned with sets on which a single binary operation has been defined. Our years of work with the integers and real numbers show that a study of sets on which two binary operations have been defined should be of great importance. Algebraic structures of this type are introduced in this section. In one sense, this section seems more intutive than those that precede it, for the structures studied are closely related to those we have worked with for many years. However, we will be continuing with our axiomatic approach. So, from another viewpoint this study is more complicated than group theory, for we now have two binary operations and more axioms to deal with.

Definitions and Basic Properties

The most general algebraic structure with two binary operations that we shall study is called a *ring*. As Example 18.2 following Definition 18.1 indicates, we have all worked with rings since grade school.

18.1 Definition A **ring** $\langle R, +, \cdot \rangle$ is a set R together with two binary operations $+$ and $\cdot$, which we call *addition* and *multiplication,* defined on R such that the following axioms are satisfied:

$\mathscr{R}_1$. $\langle R, + \rangle$ is an abelian group.

$\mathscr{R}_2$. Multiplication is associative.

$\mathscr{R}_3$. For all $a, b, c \in R$, the **left distributive law**, $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$ and the **right distributive law** $(a + b) \cdot c = (a \cdot c) + (b \cdot c)$ hold. ■

† Sections 24 and 25 are not required for the remainder of the text.

18.2 Example We are well aware that axioms $\mathscr{R}_1$, $\mathscr{R}_2$, and $\mathscr{R}_3$ for a ring hold in any subset of the complex numbers that is a group under addition and that is closed under multiplication. For example, $\langle \mathbb{Z}, +, \cdot \rangle$, $\langle \mathbb{Q}, +, \cdot \rangle$, $\langle \mathbb{R}, +, \cdot \rangle$, and $\langle \mathbb{C}, +, \cdot \rangle$ are rings. ▲

■ Historical Note

The theory of rings grew out of the study of two particular classes of rings, polynomial rings in n variables over the real or complex numbers (Section 22) and the "integers" of an algebraic number field. It was David Hilbert (1862–1943) who first introduced the term *ring,* in connection with the latter example, but it was not until the second decade of the twentieth century that a fully abstract definition appeared. The theory of commutative rings was given a firm axiomatic foundation by Emmy Noether (1882–1935) in her monumental paper "Ideal Theory in Rings," which appeared in 1921. A major concept of this paper is the ascending chain condition for ideals. Noether proved that in any ring in which every ascending chain of ideals has a maximal element, every ideal is finitely generated.

Emmy Noether received her doctorate from the University of Erlangen, Germany, in 1907. Hilbert invited her to Göttingen in 1915, but his efforts to secure her a paid position were blocked because of her sex. Hilbert complained, "I do not see that the sex of the candidate is an argument against her admission [to the faculty]. After all, we are a university, not a bathing establishment." Noether was, however, able to lecture under Hilbert's name. Ultimately, after the political changes accompanying the end of the First World War reached Göttingen, she was given in 1923 a paid position at the University. For the next decade, she was very influential in the development of the basic concepts of modern algebra. Along with other Jewish faculty members, however, she was forced to leave Göttingen in 1933. She spent the final two years of her life at Bryn Mawr College near Philadelphia.

It is customary to denote multiplication in a ring by juxtaposition, using ab in place of $a \cdot b$. We shall also observe the usual convention that multiplication is performed before addition in the absence of parentheses, so the left distributive law, for example, becomes

$$a(b + c) = ab + ac,$$

without the parentheses on the right side of the equation. Also, as a convenience analogous to our notation in group theory, we shall somewhat incorrectly refer to a *ring* R in place of a *ring* $\langle R, +, \cdot \rangle$, provided that no confusion will result. In particular, from now on $\mathbb{Z}$ will always be $\langle \mathbb{Z}, +, \cdot \rangle$, and $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ will also be the rings in Example 18.2. We may on occasion refer to $\langle R, + \rangle$ as *the additive group of the ring* R.

18.3 Example Let R be any ring and let $M_n(R)$ be the collection of all $n \times n$ matrices having elements of R as entries. The operations of addition and multiplication in R allow us to add and multiply matrices in the usual fashion, explained in the appendix. We can quickly check that $\langle M_n(R), + \rangle$ is an abelian group. The associativity of matrix multiplication and the two distributive laws in $M_n(R)$ are more tedious to demonstrate, but straightforward calculations indicate that they follow from the same properties in R. We will

assume from now on that we know that $M_n(R)$ is a ring. In particular, we have the rings $M_n(\mathbb{Z})$, $M_n(\mathbb{Q})$, $M_n(\mathbb{R})$, and $M_n(\mathbb{C})$. Note that multiplication is not a commutative operation in any of these rings for $n \geq 2$. ▲

18.4 Example Let F be the set of all functions $f : \mathbb{R} \to \mathbb{R}$. We know that $\langle F, + \rangle$ is an abelian group under the usual function addition,

$$(f + g)(x) = f(x) + g(x).$$

We define multiplication on F by

$$(fg)(x) = f(x)g(x).$$

That is, fg is the function whose value at x is $f(x)g(x)$. It is readily checked that F is a ring; we leave the demonstration to Exercise 34. We have used this juxtaposition notation $\sigma\mu$ for the composite function $\sigma(\mu(x))$ when discussing permutation multiplication. If we were to use both function multiplication and function composition in F, we would use the notation $f \circ g$ for the composite function. However, we will be using composition of functions almost exclusively with homomorphisms, which we will denote by Greek letters, and the usual product defined in this example chiefly when multiplying polynomial function $f(x)g(x)$, so no confusion should result. ▲

18.5 Example Recall that in group theory, $n\mathbb{Z}$ is the cyclic subgroup of $\mathbb{Z}$ under addition consisting of all integer multiples of the integer n. Since $(nr)(ns) = n(nrs)$, we see that $n\mathbb{Z}$, is closed under multiplication. The associative and distributive laws which hold in $\mathbb{Z}$ then assure us that $\langle n\mathbb{Z}, +, \cdot \rangle$ is as ring. From now on in the text, we will consider $n\mathbb{Z}$ to be this ring. ▲

18.6 Example Consider the cyclic group $\langle \mathbb{Z}_n, + \rangle$. If we define for $a, b \in \mathbb{Z}_n$ the product ab as the remainder of the usual product of integers when divided by n, it can be shown that $\langle \mathbb{Z}_n, +, \cdot \rangle$ is a ring. We shall feel free to use this fact. For example, in $\mathbb{Z}_{10}$ we have $(3)(7) = 1$. This operation on $\mathbb{Z}_n$ is **multiplication modulo** n. We do not check the ring axioms here, for they will follow in Section 26 from some of the theory we develop there. From now on, $\mathbb{Z}_n$ will always be the ring $\langle \mathbb{Z}_n, +, \cdot \rangle$. ▲

18.7 Example If $R_1, R_2, \cdots, R_n$ are rings, we can form the set $R_1 \times R_2 \times \cdots \times R_n$ of all ordered n-tuples $(r_1, r_2, \cdots, r_n)$, where $r_i \in R_i$. Defining addition and multiplication of n-tuples by components (just as for groups), we see at once from the ring axioms in each component that the set of all these n-tuples forms a ring under addition and multiplication by components. The ring $R_1 \times R_2 \times \cdots \times R_n$ is the **direct product** of the rings R_i. ▲

Continuing matters of notation, we shall always let 0 be the additive identity of a ring. The additive inverse of an element a of a ring is $-a$. We shall frequently have occasion to refer to a sum

$$a + a + \cdots + a$$

having n summands. We shall let this sum be $n \cdot a$, always using the dot. However, *$n \cdot a$ is not to be constructed as a multiplication of n and a in the ring, for the integer n may not be in the ring at all.* If $n < 0$, we let

$$n \cdot a = (-a) + (-a) + \cdots + (-a)$$

for $|n|$ summands. Finally, we define

$$0 \cdot a = 0$$

for $0 \in \mathbb{Z}$ on the left side of the equations and $0 \in R$ on the right side. Actually, the equation $0a = 0$ holds also for $0 \in R$ on both sides. The following theorem proves this and various other elementary but important facts. Note the strong use of the distributive laws in the proof of this theorem. Axiom $\mathscr{R}_1$ for a ring concerns only addition, and axiom $\mathscr{R}_2$ concerns only multiplication. This shows that in order to prove anything that gives a relationship between these two operations, we are going to have to use axiom $\mathscr{R}_3$. For example, the first thing that we will show in Theorem 18.8 is that $0a = 0$ for any element a in a ring R. Now this relation involves both addition and multiplication. The multiplication $0a$ stares us in the face, and 0 is an *additive* concept. Thus we will have to come up with an argument that uses a distributive law to prove this.

18.8 Theorem If R is a ring with additive identity 0, then for any $a, b \in R$ we have

1. $0a = a0 = 0$,
2. $a(-b) = (-a)b = -(ab)$,
3. $(-a)(-b) = ab$.

Proof For Property 1, note that by axioms $\mathscr{R}_1$ and $\mathscr{R}_2$,

$$a0 + a0 = a(0 + 0) = a0 = 0 + a0.$$

Then by the cancellation law for the additive group $\langle R, + \rangle$, we have $a0 = 0$. Likewise,

$$0a + 0a = (0 + 0)a = 0a = 0 + 0a$$

implies that $0a = 0$. This proves Property 1.

In order to understand the proof of Property 2, we must remember that, by *definition*, $-(ab)$ is the element that when added to ab gives 0. Thus to show that $a(-b) = -(ab)$, we must show precisely that $a(-b) + ab = 0$. By the left distributive law,

$$a(-b) + ab = a(-b + b) = a0 = 0,$$

since a0 = 0 by Property 1. Likewise,

$$(-a)b + ab = (-a + a)b = 0b = 0.$$

For Property 3, note that

$$(-a)(-b) = -(a(-b))$$

by Property 2. Again by Property 2,

$$-(a(-b)) = -(-(ab)),$$

and $-(-(ab))$ is the element that when added to $-(ab)$ gives 0. This is ab by definition of $-(ab)$ and by the uniqueness of an inverse in a group. Thus, $(-a)(-b) = ab$. ◆

It is important that you *understand* the preceding proof. The theorem allows us to use our usual rules for signs.

Homomorphisms and Isomorphisms

From our work in group theory, it is quite clear how a structure-relating map of a ring R into a ring R' should be defined.

18.9 Definition For rings R and R', a map $\phi : R \to R'$ is a **homomorphism** if the following two conditions are satisfied for all $a, b \in R$:

1. $\phi(a + b) = \phi(a) + \phi(b)$,
2. $\phi(ab) = \phi(a)\phi(b)$. ■

In the preceding definition, Condition 1 is the statement that ϕ is a homomorphism mapping the abelian group $\langle R, + \rangle$ into $\langle R', + \rangle$. Condition 2 requires that ϕ relate the multiplicative structures of the rings R and R' in the same way. Since ϕ is also a group homomorphism, all the results concerning group homomorphisms are valid for the additive structure of the rings. In particular, ϕ is one to one if and only if its **kernel** $\text{Ker}(\phi) = \{a \in R \mid \phi(a) = 0'\}$ is just the subset $\{0\}$ of R. The homomorphism ϕ of the group $\langle R, + \rangle$ gives rise to a factor group. We expect that a ring homomorphism will give rise to a factor ring. This is indeed the case. We delay discussion of this to Section 26, where the treatment will parallel our treatment of factor groups in Section 14.

18.10 Example Let F be the ring of all functions mapping $\mathbb{R}$ into $\mathbb{R}$ defined in Example 18.4. For each $a \in \mathbb{R}$, we have the **evaluation homomorphism** $\phi_a : F \to \mathbb{R}$, where $\phi_a(f) = f(a)$ for $f \in F$. We defined this homomorphism for the group $\langle F, + \rangle$ in Example 13.4, but we did not do much with it in group theory. We will be working a great deal with it in the rest of this text, for finding a real solution of a polynomial equation $p(x) = 0$ amounts precisely to finding $a \in \mathbb{R}$ such that $\phi_a(p) = 0$. Much of the remainder of this text deals with solving polynomial equations. We leave the demonstration of the multiplicative homomorphism property 2 for ϕ_a to Exercise 35. ▲

18.11 Example The map $\phi : \mathbb{Z} \to \mathbb{Z}_n$ where $\phi(a)$ is the remainder of a modulo n is a ring homomorphism for each positive integer n. We know $\phi(a + b) = \phi(a) + \phi(b)$ by group theory. To show the multiplicative property, write $a = q_1 n + r_1$ and $b = q_2 n + r_2$ according to the division algorithm. Then $ab = n(q_1 q_2 n + r_1 q_2 + q_1 r_2) + r_1 r_2$. Thus $\phi(ab)$ is the remainder of $r_1 r_2$ when divided by n. Since $\phi(a) = r_1$ and $\phi(b) = r_2$, Example 18.6 indicates that $\phi(a)\phi(b)$ is also this same remainder, so $\phi(ab) = \phi(a)\phi(b)$. From group theory, we anticipate that the ring $\mathbb{Z}_n$ might be isomorphic to a factor ring $\mathbb{Z}/n\mathbb{Z}$. This is indeed the case; factor rings will be discussed in Section 26. ▲

We realize that in the study of any sort of mathematical structure, an idea of basic importance is the concept of two systems being *structurally identical*, that is, one being just like the other except for names. In algebra this concept is always called *isomorphism*.

The concept of two things being just alike except for names of elements leads us, just as it did for groups, to the following definition.

18.12 Definition An **isomorphism** $\phi : R \to R'$ from a ring R to a ring R' is a homomorphism that is one to one and onto R'. The rings R and R' are then **isomorphic**. ■

From our work in group theory, we expect that isomorphism gives an equivalence relation on any collection of rings. We need to check that the multiplicative property of an isomorphism is satisfied for the inverse map $\phi^{-1} : R' \to R$ (to complete the symmetry argument). Similarly, we check that if $\mu : R' \to R''$ is also a ring ismorphism, then the multiplicative requirement holds for the composite map $\mu\phi : R \to R''$ (to complete the transitivity argument). We ask you to do this in Exercise 36.

18.13 Example As abelian groups, $\langle \mathbb{Z}, + \rangle$ and $\langle 2\mathbb{Z}, + \rangle$ are isomorphic under the map $\phi : \mathbb{Z} \to \mathbb{Z}$, with $\phi(x) = 2x$ for $x \in \mathbb{Z}$. Here ϕ is *not* a ring isomorphism, for $\phi(xy) = 2xy$, while $\phi(x)\phi(y) = 2x2y = 4xy$. ▲

Multiplicative Questions: Fields

Many of the rings we have mentioned, such as $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$, have a multiplicative identity element 1. However, $2\mathbb{Z}$ does not have an identity element for multiplication. Note also that multiplication is not commutative in the matrix rings described in Example 18.3.

It is evident that $\{0\}$, with $0 + 0 = 0$ and $(0)(0) = 0$, gives a ring, the **zero ring.** Here 0 acts as multiplicative as well as additive identity element. By Theorem 18.8, this is the only case in which 0 could act as a multiplicative identity element, for from $0a = 0$, we can then deduce that $a = 0$. Theorem 3.13 shows that if a ring has a multiplicative identity element, it is unique. We denote a multiplicative identity element in a ring by 1.

18.14 Definition A ring in which the multiplication is commutative is a **commutative ring.** A ring with a multiplicative identity element is a **ring with unity;** the multiplicative identity element 1 is called "**unity.**" ■

In a ring with unity 1 the distributive laws show that

$$\underbrace{(1 + 1 + \cdots + 1)}_{n \text{ summands}} \underbrace{(1 + 1 + \cdots + 1)}_{m \text{ summands}} = \underbrace{(1 + 1 + \cdots + 1)}_{nm \text{ summands}},$$

that is, $(n \cdot 1)(m \cdot 1) = (nm) \cdot 1$. The next example gives an application of this observation.

18.15 Example We claim that for integers r and s where $\gcd(r, s) = 1$, the rings $\mathbb{Z}_{rs}$ and $\mathbb{Z}_r \times \mathbb{Z}_s$ are isomorphic. Additively, they are both cyclic abelian groups of order rs with generators 1 and $(1, 1)$ respectively. Thus $\phi : \mathbb{Z}_{rs} \to \mathbb{Z}_r \times \mathbb{Z}_s$ defined by $\phi(n \cdot 1) = n \cdot (1, 1)$ is an additive group isomorphism. To check the multiplicative Condition 2 of Definition 18.9,

we use the observation preceding this example for the unity $(1, 1)$ in the ring $\mathbb{Z}_r \times \mathbb{Z}_s$, and compute.

$$\phi(nm) = (nm) \cdot (1, 1) = [n \cdot (1, 1)][m \cdot (1, 1)] = \phi(n)\phi(m). \quad \blacktriangle$$

Note that a direct product $R_1 \times R_2 \times \cdots \times R_n$ of rings is commutative or has unity if and only if each R_i is commutative or has unity, respectively.

In a ring R with unity $1 \neq 0$, the set R^* of nonzero elements, if closed under the ring multiplication, will be a multiplicative group if multiplicative inverses exist. A **multiplicative inverse** of an element a in a ring R with unity $1 \neq 0$ is an element $a^{-1} \in R$ such that $aa^{-1} = a^{-1}a = 1$. Precisely as for groups, a multiplicative inverse for an element a in R is unique, if it exists at all (see Exercise 43). Theorem 18.8 shows that it would be hopeless to have a multiplicative inverse for 0 except for the ring $\{0\}$, where $0 + 0 = 0$ and $(0)(0) = 0$, with 0 as both additive and multiplicative identity element. We are thus led to discuss the existence of multiplicative inverses for nonzero elements in a ring with nonzero unity. There is unavoidably a lot of terminology to be defined in this introductory section on rings. We are almost done.

18.16 Definition Let R be a ring with unity $1 \neq 0$. An element u in R is a **unit** of R if it has a multiplicative inverse in R. If every nonzero element of R is a unit, then R is a **division ring** (or **skew field**). A **field** is a commutative division ring. A noncommutative division ring is called a "**strictly skew field.**" ■

18.17 Example Let us find the units in $\mathbb{Z}_{14}$. Of course, 1 and $-1 = 13$ are units. Since $(3)(5) = 1$ we see that 3 and 5 are units; therefore $-3 = 11$ and $-5 = 9$ are also units. None of the remaining elements of $\mathbb{Z}_{14}$ can be units, since no multiple of 2, 4, 6, 7, 8, or 10 can be one more than a multiple of 14; they all have a common factor, either 2 or 7, with 14. Section 20 will show that the units in $\mathbb{Z}_n$ are precisely those $m \in \mathbb{Z}_n$ such that $\gcd(m, n) = 1$. ▲

18.18 Example $\mathbb{Z}$ is not a field, because 2, for example, has no multiplicative inverse, so 2 is not a unit in $\mathbb{Z}$. The only units in $\mathbb{Z}$ are 1 and -1. However, $\mathbb{Q}$ and $\mathbb{R}$ are fields. An example of a strictly skew field is given in Section 24. ▲

We have the natural concepts of a subring of a ring and subfield of a field. A **subring of a ring** is a subset of the ring that is a ring under induced operations from the whole ring; a **subfield** is defined similarly for a subset of a field. In fact, let us say here once and for all that if we have a set, together with a certain specified type of *algebraic structure* (group, ring, field, integral domain, vector space, and so on), then any subset of this set, together with a natural induced algebraic structure *that yields an algebraic structure of the same type*, is a *substructure*. If K and L are both structures, we shall let $K \leq L$ denote that K is a substructure of L and $K < L$ denote that $K \leq L$ but $K \neq L$. Exercise 48 gives criteria for a subset S of a ring R to form a subring of R.

Finally, be careful not to confuse our use of the words *unit* and *unity*. *Unity* is the multiplicative identity element, while a *unit* is any element having a multiplicative inverse. Thus the multiplicative identity element or unity is a unit, but not every unit is unity. For example, -1 is a unit in $\mathbb{Z}$, but -1 is not unity, that is, $-1 \neq 1$.

Historical Note

Although fields were implict in the early work on the solvability of equations by Abel and Galois, it was Leopold Kronecker (1823–1891) who in connection with his own work on this subject first published in 1881 a definition of what he called a "domain of rationality": "The domain of rationality $(R', R'', R''', \cdots)$ contains $\cdots$ every one of those quantities which are rational functions of the quantities $R', R'', R''', \cdots$ with integral coefficients." Kronecker, however, who insisted that any mathematical subject must be constructible in finitely many steps, did not view the domain of rationality as a complete entity, but merely as a region in which took place various operations on its elements.

Richard Dedekind (1831–1916), the inventor of the Dedekind cut definition of a real number, considered a field as a completed entity. In 1871, he published the following definition in his supplement to the second edition of Dirichlet's text on number theory: "By a field we mean any system of infinitely many real or complex numbers, which in itself is so closed and complete, that the addition, subtraction, multiplication, and division of any two numbers always produces a number of the same system." Both Kronecker and Dedekind had, however, dealt with their varying ideas of this notion as early as the 1850s in their university lectures.

A more abstract definition of a field, similar to the one in the text, was given by Heinrich Weber (1842–1913) in a paper of 1893. Weber's definition, unlike that of Dedekind, specifically included fields with finitely many elements as well as other fields, such as function fields, which were not subfields of the field of complex numbers.

EXERCISES 18

Computations

In Exercises 1 through 6, compute the product in the given ring.

1. $(12)(16)$ in $\mathbb{Z}_{24}$

2. $(16)(3)$ in $\mathbb{Z}_{32}$

3. $(11)(-4)$ in $\mathbb{Z}_{15}$

4. $(20)(-8)$ in $\mathbb{Z}_{26}$

5. $(2,3)(3,5)$ in $\mathbb{Z}_5 \times \mathbb{Z}_9$

6. $(-3,5)(2,-4)$ in $\mathbb{Z}_4 \times \mathbb{Z}_{11}$

In Exercises 7 through 13, decide whether the indicated operations of addition and multiplication are defined (closed) on the set, and give a ring structure. If a ring is not formed, tell why this is the case. If a ring is formed, state whether the ring is commutative, whether it has unity, and whether it is a field.

7. $n\mathbb{Z}$ with the usual addition and multiplication

8. $\mathbb{Z}^+$ with the usual addition and multiplication

9. $\mathbb{Z} \times \mathbb{Z}$ with addition and multiplication by components

10. $2\mathbb{Z} \times \mathbb{Z}$ with addition and multiplication by components

11. $\{a + b\sqrt{2} \mid a, b \in \mathbb{Z}\}$ with the usual addition and multiplication

12. $\{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}$ with the usual addition and multiplication

13. The set of all pure imaginary complex numbers ri for $r \in \mathbb{R}$ with the usual addition and multiplication

In Exercises 14 through 19, describe all units in the given ring

14. $\mathbb{Z}$ **15.** $\mathbb{Z} \times \mathbb{Z}$ **16.** $\mathbb{Z}_5$

17. $\mathbb{Q}$ **18.** $\mathbb{Z} \times \mathbb{Q} \times \mathbb{Z}$ **19.** $\mathbb{Z}_4$

20. Consider the matrix ring $M_2(\mathbb{Z}_2)$.

a. Find the **order** of the ring, that is, the number of elements in it.

b. List all units in the ring.

21. If possible, give an example of a homomorphism $\phi : R \to R'$ where R and R' are rings with unity $1 \neq 0$ and $1' \neq 0'$, and where $\phi(1) \neq 0'$ and $\phi(1) \neq 1'$.

22. (Linear algebra) Consider the map det of $M_n(\mathbb{R})$ into $\mathbb{R}$ where $\det(A)$ is the determinant of the matrix A for $A \in M_n(\mathbb{R})$. Is det a ring homomorphism? Why or why not?

23. Describe all ring homomorphisms of $\mathbb{Z}$ into $\mathbb{Z}$.

24. Describe all ring homomorphisms of $\mathbb{Z}$ into $\mathbb{Z} \times \mathbb{Z}$.

25. Describe all ring homomorphisms of $\mathbb{Z} \times \mathbb{Z}$ into $\mathbb{Z}$.

26. How many homomorphisms are there of $\mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$ into $\mathbb{Z}$?

27. Consider this solution of the equation $X^2 = I_3$ in the ring $M_3(\mathbb{R})$.

$X^2 = I_3$ implies $X^2 - I_3 = 0$, the zero matrix, so factoring, we have $(X - I_3)(X + I_3) = 0$

whence either $X = I_3$ or $X = -I_3$.

Is this reasoning correct? If not, point out the error, and if possible, give a counterexample to the conclusion.

28. Find all solutions of the equation $x^2 + x - 6 = 0$ in the ring $\mathbb{Z}_{14}$ by factoring the quadratic polynomial. Compare with Exercise 27.

Concepts

In Exercises 29 and 30, correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a from acceptable for publication.

29. A *field* F is a ring with nonzero unity such that the set of nonzero elements of F is a group under multiplication.

30. A *unit* in a ring is an element of magnitude 1.

31. Give an example of a ring having two elements a and b such that $ab = 0$ but neither a nor b is zero.

32. Give an example of a ring with unity $1 \neq 0$ that has a subring with nonzero unity $1' \neq 1$. [*Hint:* Consider a direct product, or a subring of $\mathbb{Z}_6$.]

33. Mark each of the following true or false.

_____ **a.** Every field is also a ring.

_____ **b.** Every ring has a multiplicative identity.

_____ **c.** Every ring with unity has at least two units.

_____ **d.** Every ring with unity has at most two units.

_____ **e.** It is possible for a subset of some field to be a ring but not a subfield, under the induced operations.
_____ **f.** The distributive laws for a ring are not very important.
_____ **g.** Multiplication in a field is commutative.
_____ **h.** The nonzero elements of a field form a group under the multiplication in the field.
_____ **i.** Addition in every ring is commutative.
_____ **j.** Every element in a ring has an additive inverse.

Theory

34. Show that the multiplication defined on the set F of functions in Example 18.4 satisfies axioms $\mathscr{R}_2$ and $\mathscr{R}_3$ for a ring.

35. Show that the evaluation map ϕ_a of Example 18.10 satisfies the multiplicative requirement for a homomorphism.

36. Complete the argument outlined after Definitions 18.12 to show that isomorphism gives an equivalence relation on a collection of rings.

37. Show that if U is the collection of all units in a ring $\langle R, +, \cdot\rangle$ with unity, then $\langle U, \cdot\rangle$ is a group. [*Warning:* Be sure to show that U is closed under multiplication.]

38. Show that $a^2 - b^2 = (a + b)(a - b)$ for all a and b in a ring R if and only if R is commutative.

39. Let $(R, +)$ be an abelian group. Show that $(R, +, \cdot)$ is a ring if we define $ab = 0$ for all $a, b \in R$.

40. Show that the rings $2\mathbb{Z}$ and $3\mathbb{Z}$ are not isomorphic. Show that the fields $\mathbb{R}$ and $\mathbb{C}$ are not isomorphic.

41. (Freshman exponentiation) Let p be a prime. Show that in the ring $\mathbb{Z}_p$ we have $(a + b)^p = a^p + b^p$ for all $a, b \in \mathbb{Z}_p$. [*Hint:* Observe that the usual binomial expansion for $(a + b)^n$ is valid in a *commutative* ring.]

42. Show that the unity element in a subfield of a field must be the unity of the whole field, in contrast to Exercise 32 for rings.

43. Show that the multiplicative inverse of a unit a ring with unity is unique.

44. An element a of a ring R is **idempotent** if $a^2 = a$.

a. Show that the set of all idempotent elements of a commutative ring is closed under multiplication.
b. Find all idempotents in the ring $\mathbb{Z}_6 \times \mathbb{Z}_{12}$.

45. (Linear algebra) Recall that for an $m \times n$ matrix A, the *transpose* A^T of A is the matrix whose jth column is the jth row of A. Show that if A is an $m \times n$ matrix such that A^TA is invertible, then the *projection matrix* $P = A(A^TA)^{-1}A^T$ is an idempotent in the ring of $n \times n$ matrices.

46. An element a of a ring R is **nilpotent** if $a^n = 0$ for some $n \in \mathbb{Z}^+$. Show that if a and b are nilpotent elements of a *commutative* ring, then $a + b$ is also nilpotent.

47. Show that a ring R has no nonzero nilpotent element if and only if 0 is the only solution of $x^2 = 0$ in R.

48. Show that a subset S of a ring R gives a subring of R if and only if the following hold:

$$0 \in S;$$
$$(a - b) \in S \text{ for all } a, b \in S;$$
$$ab \in S \text{ for all } a, b \in S.$$

49. **a.** Show that an intersection of subrings of a ring R is again a subring of R.
b. Show that an intersection of subfields of a field F is again a subfield of F.

50. Let R be a ring, and let a be a fixed element of R. Let $I_a = \{x \in R \mid ax = 0\}$. Show that I_a is a subring of R.

51. Let R be a ring, and let a be a fixed element of R. Let R_a be the subring of R that is the intersection of all subrings of R containing a (see Exercise 49). The ring R_a is the **subring of R generated by a**. Show that the abelian group $\langle R_a, + \rangle$ is generated (in the sense of Section 7) by $\{a^n \mid n \in \mathbb{Z}^+\}$.

52. (Chinese Remainder Theorem for two congruences) Let r and s be positive integers such that $\gcd(r, s) = 1$. Use the isomorphism in Example 18.15 to show that for $m, n \in \mathbb{Z}$, there exists an integer x such that $x \equiv m \pmod{r}$ and $x \equiv n \pmod{s}$.

53. **a.** State and prove the generalization of Example 18.15 for a direct product with n factors.

b. Prove the Chinese Remainder Theorem: Let $a_i, b_i \in \mathbb{Z}^+$ for $i = 1, 2, \cdots, n$ and let $\gcd(b_i, b_j) = 1$ for $i \neq j$. Then there exists $x \in \mathbb{Z}^+$ such that $x \equiv a_i \pmod{b_i}$ for $i = 1, 2, \cdots, n$.

54. Consider $\langle S, +, \cdot \rangle$, where S is a set and $+$ and $\cdot$ are binary operations on S such that

$\langle S, + \rangle$ is a group,

$\langle S^*, \cdot \rangle$ is a group where S^* consists of all elements of S except the additive identity element,

$a(b + c) = (ab) + (ac)$ and $(a + b)c = (ac) + (bc)$ for all $a, b, c \in S$.

Show that $\langle S, +, \cdot \rangle$ is a division ring. [*Hint:* Apply the distributive laws to $(1 + 1)(a + b)$ to prove the commutativity of addition.]

55. A ring R is a **Boolean ring** if $a^2 = a$ for all $a \in R$, so that every element is idempotent. Show that every Boolean ring is commutative.

56. (For students having some knowledge of the laws of set theory) For a set S, let $\mathscr{P}(S)$ be the collection of all subsets of S. Let binary operations $+$ and $\cdot$ on $\mathscr{P}(S)$ be defined by

$$A + B = (A \cup B) - (A \cap B) = \{x \mid x \in A \text{ or } x \in B \text{ but } x \notin (A \cap B)\}$$

and

$$A \cdot B = A \cap B$$

for $A, B \in \mathscr{P}(S)$.

a. Give the tables for $+$ and $\cdot$ for $\mathscr{P}(S)$, where $S = \{a, b\}$. [*Hint:* $\mathscr{P}(S)$ has four elements.]

b. Show that for *any* set S, $\langle \mathscr{P}(S), +, \cdot \rangle$ is a Boolean ring (see Exercise 55).

SECTION 19 INTEGRAL DOMAINS

While a careful treatment of polynomials is not given until Section 22, for purposes of motivation we shall make intuitive use of them in this section.

Divisors of Zero and Cancellation

One of the most important algebraic properties of our usual number system is that a product of two numbers can only be 0 if at least one of the factors is 0. We have used this fact many times in solving equations, perhaps without realizing that we were using it. Suppose, for example, we are asked to solve the equation

$$x^2 - 5x + 6 = 0.$$

The first thing we do is to factor the left side:

$$x^2 - 5x + 6 = (x - 2)(x - 3).$$

Then we conclude that the only possible values for x are 2 and 3. Why? The reason is that if x is replaced by any number a, the product $(a-2)(a-3)$ of the resulting numbers is 0 if and only if either $a-2=0$ or $a-3=0$.

19.1 Example Solve the equation $x^2-5x+6=0$ in $\mathbb{Z}_{12}$.

Solution The factorization $x^2-5x+6=(x-2)(x-3)$ is still valid if we think of x as standing for any number in $\mathbb{Z}_{12}$. But in $\mathbb{Z}_{12}$, not only is $0a=a0=0$ for all $a\in\mathbb{Z}_{12}$, but also

$$\begin{aligned}(2)(6)&=(6)(2)=(3)(4)=(4)(3)=(3)(8)=(8)(3)\\&=(4)(6)=(6)(4)=(4)(9)=(9)(4)=(6)(6)=(6)(8)\\&=(8)(6)=(6)(10)=(10)(6)=(8)(9)=(9)(8)=0.\end{aligned}$$

We find, in fact, that our equation has not only 2 and 3 as solutions, but also 6 and 11, for $(6-2)(6-3)=(4)(3)=0$ and $(11-2)(11-3)=(9)(8)=0$ in $\mathbb{Z}_{12}$. ▲

These ideas are of such importance that we formalize them in a definition.

19.2 Definition If a and b are two nonzero elements of a ring R such that $ab=0$, then a and b are **divisors of 0** (or **0 divisors**). ■

Example 19.1 shows that in $\mathbb{Z}_{12}$ the elements 2, 3, 4, 6, 8, 9, and 10 are divisors of 0. Note that these are exactly the numbers in $\mathbb{Z}_{12}$ that are not relatively prime to 12, that is, whose gcd with 12 is not 1. Our next theorem shows that this is an example of a general situation.

19.3 Theorem In the ring $\mathbb{Z}_n$, the divisors of 0 are precisely those nonzero elements that are not relatively prime to n.

Proof Let $m\in\mathbb{Z}_n$, where $m\neq 0$, and let the gcd of m and n be $d\neq 1$. Then

$$m\left(\frac{n}{d}\right)=\left(\frac{m}{d}\right)n,$$

and $(m/d)n$ gives 0 as a multiple of n. Thus $m(n/d)=0$ in $\mathbb{Z}_n$, while neither m nor n/d is 0, so m is a divisor of 0.

On the other hand, suppose $m\in\mathbb{Z}_n$ is relatively prime to n. If for $s\in\mathbb{Z}_n$ we have $ms=0$, then n divides the product ms of m and s as elements in the ring $\mathbb{Z}$. Since n is relatively prime to m, boxed Property 1 following Example 6.9 shows that n divides s, so $s=0$ in $\mathbb{Z}_n$. ◆

19.4 Corollary If p is a prime, then $\mathbb{Z}_p$ has no divisors of 0.

Proof This corollary is immediate from Theorem 19.3. ◆

Another indication of the importance of the concept of 0 divisors is shown in the following theorem. Let R be a ring, and let $a, b, c\in R$. The **cancellation laws** hold in R if $ab=ac$ with $a\neq 0$ implies $b=c$, and $ba=ca$ with $a\neq 0$ implies $b=c$. These

are multiplicative cancellation laws. Of course, the additive cancellation laws hold in R, since $\langle R, + \rangle$ is a group.

19.5 Theorem The cancellation laws hold in a ring R if and only if R has no divisors of 0.

Proof Let R be a ring in which the cancellation laws hold, and suppose $ab = 0$ for some $a, b \in R$. We must show that either a or b is 0. If $a \neq 0$, then $ab = a0$ implies that $b = 0$ by cancellation laws. Similarly, $b \neq 0$ implies that $a = 0$, so there can be no divisors of 0 if the cancellation laws hold.

Conversely, suppose that R has no divisors of 0, and suppose that $ab = ac$ with $a \neq 0$. Then

$$ab - ac = a(b - c) = 0.$$

Since $a \neq 0$, and since R has no divisors of 0, we must have $b - c = 0$, so $b = c$. A similar argument shows that $ba = ca$ with $a \neq 0$ implies $b = c$. ◆

Suppose that R is a ring with no divisors of 0. Then an equation $ax = b$, with $a \neq 0$, in R can have at most one solution x in R, for if $ax_1 = b$ and $ax_2 = b$, then $ax_1 = ax_2$, and by Theorem 19.5 $x_1 = x_2$, since R has no divisors of 0. If R has unity $1 \neq 0$ and a is a unit in R with multiplicative inverse a^{-1}, then the solution x of $ax = b$ is $a^{-1}b$. In the case that R is commutative, in particular if R is a field, it is customary to denote $a^{-1}b$ and ba^{-1} (they are equal by commutativity) by the formal quotient b/a. This quotient notation must not be used in the event that R is not commutative, for then we do not know whether b/a denotes $a^{-1}b$ or ba^{-1}. In particular, the multiplicative inverse a^{-1} of a nonzero element a of a field may be written $1/a$.

Integral Domains

The integers are really our most familiar number system. In terms of the algebraic properties we are discussing, $\mathbb{Z}$ is a commutative ring with unity and no divisors of 0. Surely this is responsible for the name that the next definition gives to such a structure.

19.6 Definition An **integral domain** D is a commutative ring with unity $1 \neq 0$ and containing no divisors of 0. ■

Thus, if the coefficients of a polynomial are from an integral domain, one can solve a polynomial equation in which the polynomial can be factored into linear factors in the usual fashion by setting each factor equal to 0.

In our hierarchy of algebraic structures, an integral domain belongs between a commutative ring with unity and a field, as we shall show. Theorem 19.5 shows that the cancellation laws for multiplication hold in an integral domain.

19.7 Example We have seen that $\mathbb{Z}$ and $\mathbb{Z}_p$ for any prime p are integral domains, but $\mathbb{Z}_n$ is not an integral domain if n is not prime. A moment of thought shows that the direct product $R \times S$ of two nonzero rings R and S is not an integral domain. Just observe that for $r \in R$ and $s \in S$ both nonzero, we have $(r, 0)(0, s) = (0, 0)$. ▲

19.8 Example Show that although $\mathbb{Z}_2$ is an integral domain, the matrix ring $M_2(\mathbb{Z}_2)$ has divisors of zero.

Solution We need only observe that

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}. \quad \blacktriangle$$

Our next theorem shows that the structure of a field is still the most restrictive (that is, the richest) one we have defined.

19.9 Theorem Every field F is an integral domain.

Proof Let $a, b \in F$, and suppose that $a \neq 0$. Then if $ab = 0$, we have

$$\left(\frac{1}{a}\right)(ab) = \left(\frac{1}{a}\right)0 = 0.$$

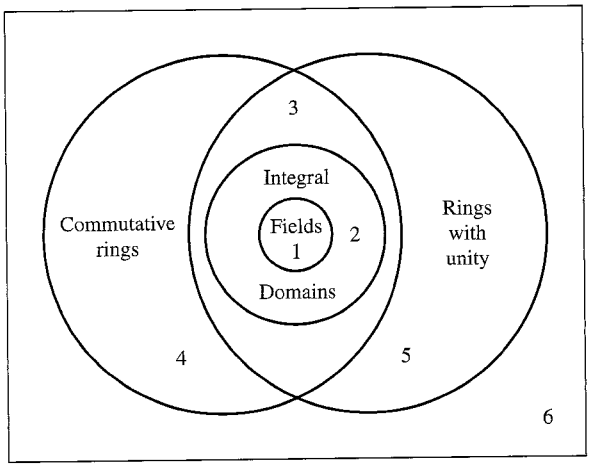
But then

$$0 = \left(\frac{1}{a}\right)(ab) = \left[\left(\frac{1}{a}\right)a\right]b = 1b = b.$$

We have shown that $ab = 0$ with $a \neq 0$ implies that $b = 0$ in F, so there are no divisors of 0 in F. Of course, F is a commutative ring with unity, so our theorem is proved. ◆

Figure 19.10 gives a Venn diagram view of containment for the algebraic structures having two binary operations with which we will be chiefly concerned. In Exercise 20 we ask you to redraw this figure to include strictly skew fields as well.

Thus far the only fields we know are $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$. The corollary of the next theorem will exhibit some fields of finite order! The proof of this theorem is a personal favorite. It is done by counting. Counting is one of the most powerful techniques in mathematics.



19.10 Figure A collection of rings.

19.11 Theorem Every finite integral domain is a field.

Proof Let

$$0, 1, a_1, \cdots, a_n$$

be all the elements of a finite domain D. We need to show that for $a \in D$, where $a \neq 0$, there exists $b \in D$ such that $ab = 1$. Now consider

$$a1, aa_1, \cdots, aa_n.$$

We claim that all these elements of D are distinct, for $aa_i = aa_j$ implies that $a_i = a_j$, by the cancellation laws that hold in an integral domain. Also, since D has no 0 divisors, none of these elements is 0. Hence by *counting,* we find that $a1, aa_1, \cdots, aa_n$ are elements $1, a_1, \cdots, a_n$ in some order, so that either $a1 = 1$, that is, $a = 1$, or $aa_i = 1$ for some i. Thus a has a multiplicative inverse. ◆

19.12 Corollary If p is a prime, then $\mathbb{Z}_p$ is a field.

Proof This corollary follows immediately from the fact that $\mathbb{Z}_p$ is an integral domain and from Theorem 19.11. ◆

The preceding corollary shows that when we consider the ring $M_n(\mathbb{Z}_p)$, we are talking about a ring of matrices over a *field.* In the typical undergraduate linear algebra course, only the field properties of the real or complex numbers are used in much of the work. Such notions as matrix reduction to solve linear systems, determinants, Cramer's rule, eigenvalues and eigenvectors, and similarity transformations to try to diagonalize a matrix are valid using matrices over any field; they depend only on the arithmetic properties of a field. Considerations of linear algebra involving notions of magnitude, such as least-squares approximate solutions or orthonormal bases, only make sense using fields where we have an idea of magnitude. The relation

$$p \cdot 1 = \underbrace{1 + 1 + \cdots + 1}_{p \text{ summands}} = 0$$

indicates that there can be no very natural notion of magnitude in the field $\mathbb{Z}_p$.

The Characteristic of a Ring

Let R be any ring. We might ask whether there is a positive integer n such that $n \cdot a = 0$ for all $a \in R$, where $n \cdot a$ means $a + a + \cdots + a$ for n summands, as explained in Section 18. For example, the integer m has this property for the ring $\mathbb{Z}_m$.

19.13 Definition If for a ring R a positive integer n exists such that $n \cdot a = 0$ for all $a \in R$, then the least such positive integer is the **characteristic of the ring** R. If no such positive integer exists, then R is of **characteristic** 0. ■

We shall be using the concept of a characteristic chiefly for fields. Exercise 29 asks us to show that the characteristic of an integral domain is either 0 or a prime p.

19.14 Example The ring $\mathbb{Z}_n$ is of characteristic n, while $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ all have characteristic 0. ▲

At first glance, determination of the characteristic of a ring seems to be a tough job, unless the ring is obviously of characteristic 0. Do we have to examine *every* element a of the ring in accordance with Definition 19.13? Our final theorem of this section shows that if the ring has unity, it suffices to examine only $a = 1$.

19.15 Theorem Let R be a ring with unity. If $n \cdot 1 \neq 0$ for all $n \in \mathbb{Z}^+$, then R has characteristic 0. If $n \cdot 1 = 0$ for some $n \in \mathbb{Z}^+$, then the smallest such integer n is the characteristic of R.

Proof If $n \cdot 1 \neq 0$ for all $n \in \mathbb{Z}^+$, then surely we cannot have $n \cdot a = 0$ for all $a \in R$ for some positive integer n, so by Definition 19.13, R has characteristic 0.

Suppose that n is a positive integer such that $n \cdot 1 = 0$. Then for any $a \in R$, we have

$$n \cdot a = a + a + \cdots + a = a(1 + 1 + \cdots + 1) = a(n \cdot 1) = a0 = 0.$$

Our theorem follows directly. ◆

EXERCISES 19

Computations

1. Find all solutions of the equation $x^3 - 2x^2 - 3x = 0$ in $\mathbb{Z}_{12}$.

2. Solve the equation $3x = 2$ in the field $\mathbb{Z}_7$; in the field $\mathbb{Z}_{23}$.

3. Find all solutions of the equation $x^2 + 2x + 2 = 0$ in $\mathbb{Z}_6$.

4. Find all solutions of $x^2 + 2x + 4 = 0$ in $\mathbb{Z}_6$.

In Exercises 5 through 10, find the characteristic of the given ring.

5. $2\mathbb{Z}$

6. $\mathbb{Z} \times \mathbb{Z}$

7. $\mathbb{Z}_3 \times 3\mathbb{Z}$

8. $\mathbb{Z}_3 \times \mathbb{Z}_3$

9. $\mathbb{Z}_3 \times \mathbb{Z}_4$

10. $\mathbb{Z}_6 \times \mathbb{Z}_{15}$

11. Let R be a commutative ring with unity of characteristic 4. Compute and simplify $(a + b)^4$ for $a, b \in R$.

12. Let R be a commutative ring with unity of characteristic 3. Compute and simplify $(a + b)^9$ for $a, b \in R$.

13. Let R be a commutative ring with unity of characteristic 3. Compute and simplify $(a + b)^6$ for $a, b \in R$.

14. Show that the matrix $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ is a divisor of zero in $M_2(\mathbb{Z})$.

Concepts

In Exercises 15 and 16, correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a form acceptable for publication.

15. If $ab = 0$, then a and b are *divisors of zero*.

16. If $n \cdot a = 0$ for all elements a in a ring R, then n is the *characteristic of* R.

17. Mark each of the following true or false.

_____ **a.** $n\mathbb{Z}$ has zero divisors if n is not prime.

_____ **b.** Every field is an integral domain.

_____ **c.** The characteristic of $n\mathbb{Z}$ is n.

_____ **d.** As a ring, $\mathbb{Z}$ is isomorphic to $n\mathbb{Z}$ for all $n \geq 1$.

_____ **e.** The cancellation law holds in any ring that is isomorphic to an integral domain.

_____ **f.** Every integral domain of characteristic 0 is infinite.

_____ **g.** The direct product of two integral domains is again an integral domain.

_____ **h.** A divisor of zero in a commutative ring with unity can have no multiplicative inverse.

_____ **i.** $n\mathbb{Z}$ is a subdomain of $\mathbb{Z}$.

_____ **j.** $\mathbb{Z}$ is a subfield of $\mathbb{Q}$.

18. Each of the six numbered regions in Fig. 19.10 corresponds to a certain type of a ring. Give an example of a ring in each of the six cells. For example, a ring in the region numbered 3 must be commutative (it is inside the commutative circle), have unity, but not be an integral domain.

19. (For students who have had a semester of linear algebra) Let F be a field. Give five different characterizations of the elements A of $M_n(F)$ that are divisors of 0.

20. Redraw Fig. 19.10 to include a subset corresponding to strictly skew fields.

Proof Synopsis

21. Give a one-sentence synopsis of the proof of the "if" part of Theorem 19.5.

22. Give a one-sentence synopsis of the proof of Theorem 19.11.

Theory

23. An element a of a ring R is **idempotent** if $a^2 = a$. Show that a division ring contains exactly two idempotent elements.

24. Show that an intersection of subdomains of an integral domain D is again a subdomain of D.

25. Show that a finite ring R with unity $1 \neq 0$ and no divisors of 0 is a division ring. (It is actually a field, although commutativity is not easy to prove. See Theorem 24.10.) [*Note:* In your proof, to show that $a \neq 0$ is a unit, you must show that a "left multiplicative inverse" of $a \neq 0$ in R is also a "right multiplicative inverse."]

26. Let R be a ring that contains at least two elements. Suppose for each nonzero $a \in R$, there exists a unique $b \in R$ such that $aba = a$.

a. Show that R has no divisors of 0.

b. Show that $bab = b$.

c. Show that R has unity.

d. Show that R is a division ring.

27. Show that the characteristic of a subdomain of an integral domain D is equal to the characteristic of D.

28. Show that if D is an integral domain, then $\{n \cdot 1 \mid n \in \mathbb{Z}\}$ is a subdomain of D contained in every subdomain of D.

29. Show that the characteristic of an integral domain D must be either 0 or a prime p. [*Hint:* If the characteristic of D is mn, consider $(m \cdot 1)(n \cdot 1)$ in D.]

30. This exercise shows that every ring R can be enlarged (if necessary) to a ring S with unity, having the same characteristic as R. Let $S = R \times Z$ if R has characteristic 0, and $R \times \mathbb{Z}_n$ if R has characteristic n. Let addition in S be the usual addition by components, and let multiplication be defined by

$$(r_1, n_1)(r_2, n_2) = (r_1 r_2 + n_1 \cdot r_2 + n_2 \cdot r_1, n_1 n_2)$$

where $n \cdot r$ has the meaning explained in Section 18.

a. Show that S is a ring.
b. Show that S has unity.
c. Show that S and R have the same characteristic.
d. Show that the map $\phi : R \to S$ given by $\phi(r) = (r, 0)$ for $r \in R$ maps R isomorphically onto a subring of S.

SECTION 20 FERMAT'S AND EULER'S THEOREMS

Fermat's Theorem

We know that as additive groups, $\mathbb{Z}_n$ and $\mathbb{Z}/n\mathbb{Z}$ are naturally isomorphic, with the coset $a + n\mathbb{Z}$ corresponding to a for each $a \in \mathbb{Z}_n$. Furthermore, addition of cosets in $\mathbb{Z}/n\mathbb{Z}$ may be performed by choosing any representatives, adding them in $\mathbb{Z}$, and finding the coset of $n\mathbb{Z}$ containing their sum. It is easy to see that $\mathbb{Z}/n\mathbb{Z}$ can be made into a ring by multiplying cosets in the same fashion, that is, by multiplying any chosen representatives. While we will be showing this later in a more general situation, we do this special case now. We need only show that such coset multiplication is well defined, because the associativity of multiplication and the distributive laws will follow immediately from those properties of the chosen representatives in $\mathbb{Z}$. To this end, choose representatives $a + rn$ and $b + sn$, rather than a and b, from the cosets $a + n\mathbb{Z}$ and $b + n\mathbb{Z}$. Then

$$(a + rn)(b + sn) = ab + (as + rb + rsn)n,$$

which is also an element of $ab + n\mathbb{Z}$. Thus the multiplication is well-defined, and our cosets form a ring isomorphic to the ring $\mathbb{Z}_n$.

The following is a special case of Exercise 37 in Section 18.

> For any field, the nonzero elements form a group under the field multiplication.

In particular, for $\mathbb{Z}_p$, the elements

$$1, 2, 3, \cdots, p - 1$$

form a group of order $p - 1$ under multiplication modulo p. Since the order of any element in a group divides the order of the group, we see that for $b \neq 0$ and $b \in \mathbb{Z}_p$, we have $b^{p-1} = 1$ in $\mathbb{Z}_p$. Using the fact that $\mathbb{Z}_p$ is isomorphic to the ring of cosets of the form $a + p\mathbb{Z}$ described above, we see at once that for any $a \in \mathbb{Z}$ not in the coset $0 + p\mathbb{Z}$, we must have

$$a^{p-1} \equiv 1 \pmod{p}.$$

This gives us at once the so-called Little Theorem of Fermat.

20.1 Theorem **(Little Theorem of Fermat)** If $a \in \mathbb{Z}$ and p is a prime not dividing a, then p divides $a^{p-1} - 1$, that is, $a^{p-1} \equiv 1 \pmod{p}$ for $a \not\equiv 0 \pmod{p}$.

20.2 Corollary If $a \in \mathbb{Z}$, then $a^p \equiv a \pmod{p}$ for any prime p.

Proof The corollary follows from Theorem 20.1 if $a \not\equiv 0 \pmod{p}$. If $a \equiv 0 \pmod{p}$, then both sides reduce to 0 modulo p. ◆

Historical Note

The statement of Theorem 20.1 occurs in a letter from Pierre de Fermat (1601–1665) to Bernard Frenicle de Bessy, dated 18 October 1640. Fermat's version of the theorem was that for any prime p and any geometric progression $a, a^2, \cdots, a^t, \cdots$, there is a least number a^T of the progression such that p divides $a^T - 1$. Furthermore, T divides $p - 1$ and p also divides all numbers of the form $a^{KT} - 1$. (It is curious that Fermat failed to note the condition that p not divide a; perhaps he felt that it was obvious that the result fails in that case.)

Fermat did not in the letter or elsewhere indicate a proof of the result and, in fact, never mentioned it again. But we can infer from other parts of this correspondence that Fermat's interest in this result came from his study of perfect numbers. (A perfect number is a positive integer m that is the sum of all of its divisors less than m; for example, $6 = 1 + 2 + 3$ is a perfect number.) Euclid had shown that $2^{n-1}(2^n - 1)$ is perfect if $2^n - 1$ is prime. The question then was to find methods for determining whether $2^n - 1$ was prime. Fermat noted that $2^n - 1$ was composite if n is composite, and then derived from his theorem the result that if n is prime, the only possible divisors of $2^n - 1$ are those of the form $2kn + 1$. From this result he was able quickly to show, for example, that $2^{37} - 1$ was divisible by $223 = 2 \cdot 3 \cdot 37 + 1$.

20.3 Example Let us compute the remainder of 8^{103} when divided by 13. Using Fermat's theorem, we have

$$\begin{aligned} 8^{103} &\equiv (8^{12})^8(8^7) \equiv (1^8)(8^7) \equiv 8^7 \equiv (-5)^7 \\ &\equiv (25)^3(-5) \equiv (-1)^3(-5) \equiv 5 \pmod{13}. \end{aligned}$$ ▲

20.4 Example Show that $2^{11,213} - 1$ is not divisible by 11.

Solution By Fermat's theorem, $2^{10} \equiv 1 \pmod{11}$, so

$$\begin{aligned} 2^{11,213} - 1 &\equiv [(2^{10})^{1,121} \cdot 2^3] - 1 \equiv [1^{1,121} \cdot 2^3] - 1 \\ &\equiv 2^3 - 1 \equiv 8 - 1 \equiv 7 \pmod{11}. \end{aligned}$$

Thus the remainder of $2^{11,213} - 1$ when divided by 11 is 7, not 0. (The number 11,213 is prime, and it has been shown that $2^{11,213} - 1$ is a prime number. Primes of the form $2^p - 1$ where p is prime are known as **Mersenne primes.**) ▲

20.5 Example Show that for every integer n, the number $n^{33} - n$ is divisible by 15.

Solution This seems like an incredible result. It means that 15 divides $2^{33} - 2$, $3^{33} - 3$, $4^{33} - 4$, etc.

Now $15 = 3 \cdot 5$, and we shall use Fermat's theorem to show that $n^{33} - n$ is divisible by both 3 and 5 for every n. Note that $n^{33} - n = n(n^{32} - 1)$.

If 3 divides n, then surely 3 divides $n(n^{32} - 1)$. If 3 does not divide n, then by Fermat's theorem, $n^2 \equiv 1 \pmod 3$ so

$$n^{32} - 1 \equiv (n^2)^{16} - 1 \equiv 1^{16} - 1 \equiv 0 \pmod 3,$$

and hence 3 divides $n^{32} - 1$.

If $n \equiv 0 \pmod 5$, then $n^{33} - n \equiv 0 \pmod 5$. If $n \not\equiv 0 \pmod 5$, then by Fermat's theorem, $n^4 \equiv 1 \pmod 5$, so

$$n^{32} - 1 \equiv (n^4)^8 - 1 \equiv 1^8 - 1 \equiv 0 \pmod 5.$$

Thus $n^{33} - n \equiv 0 \pmod 5$ for every n also. ▲

Euler's Generalization

Euler gave a generalization of Fermat's theorem. His generalization will follow at once from our next theorem, which is proved by *counting*, using essentially the same argument as in Theorem 19.11.

20.6 Theorem The set G_n of nonzero elements of $\mathbb{Z}_n$ that are not 0 divisors forms a group under multiplication modulo n.

Proof First we must show that G_n is closed under multiplication modulo n. Let $a, b \in G_n$. If $ab \notin G_n$, then there would exist $c \neq 0$ in $\mathbb{Z}_n$ such that $(ab)c = 0$. Now $(ab)c = 0$ implies that $a(bc) = 0$. Since $b \in G_n$ and $c \neq 0$, we have $bc \neq 0$ by definition of G_n. But then $a(bc) = 0$ would imply that $a \notin G_n$ contrary to assumption. *Note that we have shown that for any ring the set of elements that are not divisors of* 0 *is closed under multiplication.* No structure of $\mathbb{Z}_n$ other than ring structure has been involved so far.

We now show that G_n is a group. Of course, multiplication modulo n is associative, and $1 \in G_n$. It remains to show that for $a \in G_n$, there is $b \in G_n$ such that $ab = 1$. Let

$$1, a_1, \cdots, a_r$$

be the elements of G_n. The elements

$$a1, aa_1, \cdots, aa_r$$

are all different, for if $aa_i = aa_j$, then $a(a_i - a_j) = 0$, and since $a \in G_n$ and thus is not a divisor of 0, we must have $a_i - a_j = 0$ or $a_i = a_j$. Therefore by counting, we find that either $a1 = 1$, or some aa_i must be 1, so a has a multiplicative inverse. ◆

Note that the only property of $\mathbb{Z}_n$ used in this last theorem, other than the fact that it was a ring with unity, was that it was finite. In both Theorem 19.11 and Theorem 20.6 we have (in essentially the same construction) employed a counting argument. *Counting arguments are often simple, but they are among the most powerful tools of mathematics.*

Let n be a positive integer. Let $\varphi(n)$ be defined as the number of positive integers less than or equal to n and relatively prime to n. Note that $\varphi(1) = 1$.

20.7 Example Let $n = 12$. The positive integers less than or equal to 12 and relatively prime to 12 are 1, 5, 7, and 11, so $\varphi(12) = 4$. ▲

By Theorem 19.3, $\varphi(n)$ is the number of nonzero elements of $\mathbb{Z}_n$ that are not divisors of 0. This function $\varphi : \mathbb{Z}^+ \to \mathbb{Z}^+$ is the **Euler phi-function.** We can now describe Euler's generalization of Fermat's theorem.

20.8 Theorem **(Euler's Theorem)** If a is an integer relatively prime to n, then $a^{\varphi(n)} - 1$ is divisible by n, that is, $a^{\varphi(n)} \equiv 1 \pmod{n}$.

Proof If a is relatively prime to n, then the coset $a + n\mathbb{Z}$ of $n\mathbb{Z}$ containing a contains an integer $b < n$ and relatively prime to n. Using the fact that multiplication of these cosets by multiplication modulo n of representatives is well-defined, we have

$$a^{\varphi(n)} \equiv b^{\varphi(n)} \pmod{n}.$$

But by Theorems 19.3 and 20.6, b can be viewed as an element of the multiplicative group G_n of order $\varphi(n)$ consisting of the $\varphi(n)$ elements of $\mathbb{Z}_n$ relatively prime to n. Thus

$$b^{\varphi(n)} \equiv 1 \pmod{n},$$

and our theorem follows. ◆

20.9 Example Let $n = 12$. We saw in Example 20.7 that $\varphi(12) = 4$. Thus if we take any integer a relatively prime to 12, then $a^4 \equiv 1 \pmod{12}$. For example, with $a = 7$, we have $7^4 = (49)^2 = 2,401 = 12(200) + 1$, so $7^4 \equiv 1 \pmod{12}$. Of course, the easy way to compute $7^4 \pmod{12}$, without using Euler's theorem, is to compute it in $\mathbb{Z}_{12}$. In $\mathbb{Z}_{12}$, we have $7 = -5$ so

$$7^2 = (-5)^2 = (5)^2 = 1 \qquad \text{and} \qquad 7^4 = 1^2 = 1.$$ ▲

Application to $ax \equiv b \pmod{m}$

Using Theorem 20.6, we can find all solutions of a linear congruence $ax \equiv b \pmod{m}$. We prefer to work with an equation in $\mathbb{Z}_m$ and interpret the results for congruences.

20.10 Theorem Let m be a positive integer and let $a \in \mathbb{Z}_m$ be relatively prime to m. For each $b \in \mathbb{Z}_m$, the equation $ax = b$ has a unique solution in $\mathbb{Z}_m$.

Proof By Theorem 20.6, a is a unit in $\mathbb{Z}_m$ and $s = a^{-1}b$ is certainly a solution of the equation. Multiplying both sides of $ax = b$ on the left by a^{-1}, we see this is the only solution. ◆

Interpreting this theorem for congruences, we obtain at once the following corollary.

20.11 Corollary If a and m are relatively prime intergers, then for any integer b, the congruence $ax \equiv b \pmod{m}$ has as solutions all integers in precisely one residue class modulo m.

Theorem 20.10 serves as a lemma for the general case.

20.12 Theorem Let m be a positive integer and let $a, b \in \mathbb{Z}_m$. Let d be the gcd of a and m. The equation $ax = b$ has a solution in $\mathbb{Z}_m$ if and only if d divides b. When d divides b, the equation has exactly d solutions in $\mathbb{Z}_m$.

Proof First we show there is no solution of $ax = b$ in $\mathbb{Z}_m$ unless d divides b. Suppose $s \in \mathbb{Z}_m$ is a solution. Then $as - b = qm$ in $\mathbb{Z}$, so $b = as - qm$. Since d divides both a and m, we see that d divides the right-hand side of the equation $b = as - qm$, and hence divides b. Thus a solution s can exist only if d divides b.

Suppose now that d does divide b. Let

$$a = a_1 d, \quad b = b_1 d, \quad \text{and} \quad m = m_1 d.$$

Then the equation $as - b = qm$ in $\mathbb{Z}$ can be rewritten as $d(a_1 s - b_1) = dqm_1$. We see that $as - b$ is a multiple of m if and only if $a_1 s - b_1$ is a multiple of m_1. Thus the solutions s of $ax = b$ in $\mathbb{Z}_m$ are precisely the elements that, read modulo m_1, yield solutions of $a_1 x = b_1$ in $\mathbb{Z}_{m_1}$. Now let $s \in \mathbb{Z}_{m_1}$ be the unique solution of $a_1 x = b_1$ in $\mathbb{Z}_{m_1}$ given by Theorem 20.10. The numbers in $\mathbb{Z}_m$ that reduce to s modulo m_1 are precisely those that can be computed in $\mathbb{Z}_m$ as

$$s, s + m_1, s + 2m_1, s + 3m_1, \cdots, s + (d-1)m_1.$$

Thus there are exactly d solutions of the equation in $\mathbb{Z}_m$. ◆

Theorem 20.12 gives us at once this classical result on the solutions of a linear congruence.

20.13 Corollary Let d be the gcd of positive integers a and m. The congruence $ax \equiv b \pmod{m}$ has a solution if and only if d divides b. When this is the case, the solutions are the integers in exactly d distinct residue classes modulo m.

Actually, our proof of Theorem 20.12 shows a bit more about the solutions of $ax \equiv b \pmod{m}$ than we stated in this corollary; namely, it shows that if any solution s is found, then the solutions are precisely all elements of the residue classes $(s + km_1) + (m\mathbb{Z})$ where $m_1 = m/d$ and k runs through the integers from 0 to $d - 1$. It also tells us that we can find such an s by finding $a_1 = a/d$ and $b_1 = b/d$, and solving $a_1 x \equiv b_1 \pmod{m_1}$. To solve this congruence, we may consider a_1 and b_1 to be replaced by their remainders modulo m_1 and solve the equation $a_1 x = b_1$ in $\mathbb{Z}_{m_1}$.

20.14 Example Find all solutions of the congruence $12x \equiv 27 \pmod{18}$.

Solution The gcd of 12 and 18 is 6, and 6 is not a divisor of 27. Thus by the preceding corollary, there are no solutions. ▲

20.15 Example Find all solutions of the congruence $15x \equiv 27 \pmod{18}$.

Solution The gcd of 15 and 18 is 3, and 3 does divide 27. Proceeding as explained before Example 20.14, we divide everything by 3 and consider the congruence $5x \equiv 9 \pmod{6}$, which amounts to solving the equation $5x = 3$ in $\mathbb{Z}_6$. Now the units in $\mathbb{Z}_6$ are 1 and 5, and 5 is clearly its own inverse in this group of units. Thus the solution in $\mathbb{Z}_6$ is $x = (5^{-1})(3) = (5)(3) = 3$. Consequently, the solutions of $15x \equiv 27 \pmod{18}$ are the integers in the three residue classes.

$$3 + 18\mathbb{Z} = \{\cdots, -33, -15, 3, 21, 39, \cdots\},$$
$$9 + 18\mathbb{Z} = \{\cdots, -27, -9, 9, 27, 45, \cdots\}.$$
$$15 + 18\mathbb{Z} = \{\cdots, -21, -3, 15, 33, 51, \cdots\},$$

illustrating Corollary 20.13. Note the $d = 3$ solutions 3, 9, and 15 in $\mathbb{Z}_{18}$. All the solutions in the three displayed residue classes modulo 18 can be collected in the one residue class $3 + 6\mathbb{Z}$ modulo 6, for they came from the solution $x = 3$ of $5x = 3$ in $\mathbb{Z}_6$. ▲

EXERCISES 20

Computations

We will see later that the multiplicative group of nonzero elements of a finite field is cyclic. Illustrate this by finding a generator for this group for the given finite field.

1. $\mathbb{Z}_7$ **2.** $\mathbb{Z}_{11}$ **3.** $\mathbb{Z}_{17}$

4. Using Fermat's theorem, find the remainder of 3^{47} when it is divided by 23.

5. Use Fermat's theorem to find the remainder of 37^{49} when it is divided by 7.

6. Compute the remainder of $2^{(2^{17})} + 1$ when divided by 19. [*Hint:* You will need to compute the remainder of 2^{17} modulo 18.]

7. Make a table of values of $\varphi(n)$ for $n \leq 30$.

8. Compute $\varphi(p^2)$ where p is a prime.

9. Compute $\varphi(pq)$ where both p and q are primes.

10. Use Euler's generalization of Fermat's theorem to find the remainder of 7^{1000} when divided by 24.

In Exercises 11 through 18, describe all solutions of the given congruence, as we did in Examples 20.14 and 20.15.

11. $2x \equiv 6 \pmod 4$

12. $22x \equiv 5 \pmod{15}$

13. $36x \equiv 15 \pmod{24}$

14. $45x \equiv 15 \pmod{24}$

15. $39x \equiv 125 \pmod 9$

16. $41x \equiv 125 \pmod 9$

17. $155x \equiv 75 \pmod{65}$

18. $39x \equiv 52 \pmod{130}$

19. Let p be a prime ≥ 3. Use Exercise 28 below to find the remainder of $(p-2)!$ modulo p.

20. Using Exercise 28 below, find the remainder of 34! modulo 37.

21. Using Exercise 28 below, find the remainder of 49! modulo 53.

22. Using Exercise 28 below, find the remainder of 24! modulo 29.

Concepts

23. Mark each of the following true or false.

_____ **a.** $a^{p-1} \equiv 1 \pmod p$ for all integers a and primes p.

_____ **b.** $a^{p-1} \equiv 1 \pmod p$ for all integers a such that $a \not\equiv 0 \pmod p$ for a prime p.

_____ **c.** $\varphi(n) \leq n$ for all $n \in \mathbb{Z}^+$.

_____ **d.** $\varphi(n) \leq n - 1$ for all $n \in \mathbb{Z}^+$.

_____ **e.** The units in $\mathbb{Z}_n$ are the positive integers less than n and relatively prime to n.

_____ **f.** The product of two units in $\mathbb{Z}_n$ is always a unit.

_____ **g.** The product of two nonunits in $\mathbb{Z}_n$ may be a unit.

_____ **h.** The product of a unit and a nonunit in $\mathbb{Z}_n$ is never a unit.

_____ **i.** Every congruence $ax \equiv b \pmod{p}$, where p is a prime, has a solution.

_____ **j.** Let d be the gcd of positive integers a and m. If d divides b, then the congruence $ax \equiv b \pmod{m}$ has exactly d incongruent solutions.

24. Give the group multiplication table for the multiplicative group of units in $\mathbb{Z}_{12}$. To which group of order 4 is it isomorphic?

Proof Synopsis

25. Give a one-sentence synopsis of the proof of Theorem 20.1.

26. Give a one-sentence synopsis of the proof of Theorem 20.8.

Theory

27. Show that 1 and $p - 1$ are the only elements of the field $\mathbb{Z}_p$ that are their own multiplicative inverse. [*Hint:* Consider the equation $x^2 - 1 = 0$.]

28. Using Exercise 27, deduce the half of *Wilson's theorem* that states that if p is a prime, then $(p - 1)! \equiv -1 \pmod{p}$. [The other half states that if n is an integer >1 such that $(n - 1)! \equiv -1 \pmod{n}$, then n is a prime. Just think what the remainder of $(n - 1)!$ would be modulo n if n is not a prime.]

29. Use Fermat's theorem to show that for any positive integer n, the integer $n^{37} - n$ is divisible by 383838. [*Hint:* $383838 = (37)(19)(13)(7)(3)(2)$.]

30. Referring to Exercise 29, find a number larger than 383838 that divides $n^{37} - n$ for all positive integers n.

SECTION 21 The Field of Quotients of an Integral Domain

If an integral domain is such that every nonzero element has a multiplicative inverse, then it is a field. However, many integral domains, such as the integers $\mathbb{Z}$, do not form a field. This dilemma is not too serious. It is the purpose of this section to show that every integral domain can be regarded as being contained in a certain field, *a field of quotients of the integral domain*. This field will be a minimal field containing the integral domain in a sense that we shall describe. For example, the integers are contained in the field $\mathbb{Q}$, whose elements can all be expressed as quotients of integers. Our construction of a field of quotients of an integral domain is exactly the same as the construction of the rational numbers from the integers, which often appears in a course in foundations or advanced calculus. To follow this construction through is such a good exercise in the use of definitions and the concept of isomorphism that we discuss it in some detail, although to write out, or to read, every last detail would be tedious. We can be motivated at every step by the way $\mathbb{Q}$ can be formed from $\mathbb{Z}$.

The Construction

Let D be an integral domain that we desire to enlarge to a field of quotients F. A coarse outline of the steps we take is as follows:

1. Define what the elements of F are to be.
2. Define the binary operations of addition and multiplication on F.

3. Check all the field axioms to show that F is a field under these operations.
4. Show that F can be viewed as containing D as an integral subdomain.

Steps 1, 2, and 4 are very interesting, and Step 3 is largely a mechanical chore. We proceed with the construction.

Step 1 Let D be a given integral domain, and form the Cartesian product

$$D \times D = \{(a, b) \mid a, b \in D\}$$

We are going to think of an ordered pair (a, b) as representing a *formal quotient* a/b, that is, if $D = \mathbb{Z}$, the pair (2, 3) will eventually represent the number $\frac{2}{3}$ for us. The pair (2, 0) represents no element of $\mathbb{Q}$ and suggests that we cut the set $D \times D$ down a bit. Let S be the subset of $D \times D$ given by

$$S = \{(a, b) \mid a, b \in D, b \neq 0\}.$$

Now S is still not going to be our field as is indicated by the fact that, with $D = \mathbb{Z}$, *different* pairs of integers such as (2, 3) and (4, 6) can represent the *same* rational number. We next define when two elements of S will eventually represent the same element of F, or, as we shall say, when two elements of S are *equivalent*.

21.1 Definition Two elements (a, b) and (c, d) in S are **equivalent,** denoted by $(a, b) \sim (c, d)$, if and only if $ad = bc$. ■

Observe that this definition is reasonable, since the criterion for $(a, b) \sim (c, d)$ is an equation $ad = bc$ involving elements in D and concerning the known multiplication in D. Note also that for $D = \mathbb{Z}$, the criterion gives us our usual definition of *equality* of $\frac{a}{b}$ with $\frac{c}{d}$, for example, $\frac{2}{3} = \frac{4}{6}$ since (2)(6) = (3)(4). The rational number that we usually denote by $\frac{2}{3}$ can be thought of as the collection of *all* quotients of integers that reduce to, or are equivalent to, $\frac{2}{3}$.

21.2 Lemma The relation $\sim$ between elements of the set S as just described is an equivalence relation.

Proof We must check the three properties of an equivalence relation.

Reflexive $(a, b) \sim (a, b)$ since $ab = ba$, for multiplication in D is commutative.

Symmetric If $(a, b) \sim (c, d)$, then $ad = bc$. Since multiplication in D is commutative, we deduce that $cb = da$, and consequently $(c, d) \sim (a, b)$.

Transitive If $(a, b) \sim (c, d)$ and $(c, d) \sim (r, s)$, then $ad = bc$ and $cs = dr$. Using these relations and the fact that multiplication in D is commutative, we have

$$asd = sad = sbc = bcs = bdr = brd.$$

Now $d \neq 0$, and D is an integral domain, so cancellation is valid; this is a crucial step in the argument. Hence from $asd = brd$ we obtain $as = br$, so that $(a, b) \sim (r, s)$.

◆

We now know, in view of Theorem 0.22, that $\sim$ gives a partition of S into equivalence classes. To avoid long bars over extended expressions, we shall let $[(a, b)]$, rather than $\overline{(a, b)}$, be the equivalence class of (a, b) in S under the relation $\sim$. We now finish Step 1 by defining F to be the set of all equivalence classes $[(a, b)]$ for $(a, b) \in S$.

Step 2 The next lemma serves to define addition and multiplication in F. Observe that if $D = \mathbb{Z}$ and $[(a, b)]$ is viewed as $(a/b) \in \mathbb{Q}$, these definitions applied to $\mathbb{Q}$ give the usual operations.

21.3 Lemma For $[(a, b)]$ and $[(c, d)]$ in F, the equations

$$[(a, b)] + [(c, d)] = [(ad + bc, bd)]$$

and

$$[(a, b)][(c, d)] = [(ac, bd)]$$

give well-defined operations of addition and multiplication on F.

Proof Observe first that if $[(a, b)]$ and $[(c, d)]$ are in F, then (a, b) and (c, d) are in S, so $b \neq 0$ and $d \neq 0$. Because D is an integral domain, $bd \neq 0$, so both $(ad + bc, bd)$ and (ac, bd) are in S. (Note the crucial use here of the fact that D has no divisors of 0.) This shows that the right-hand sides of the defining equations are at least in F.

It remains for us to show that these operations of addition and multiplication are well defined. That is, they were defined by means of representatives in S of elements of F, so we must show that if different representatives in S are chosen, the same element of F will result. To this end, suppose that $(a_1, b_1) \in [(a, b)]$ and $(c_1, d_1) \in [(c, d)]$. We must show that

$$(a_1d_1 + b_1c_1, b_1d_1) \in [(ad + bc, bd)]$$

and

$$(a_1c_1, b_1d_1) \in [(ac, bd)].$$

Now $(a_1, b_1) \in [(a, b)]$ means that $(a_1, b_1) \sim (a, b)$; that is,

$$a_1b = b_1a.$$

Similarly, $(c_1, d_1) \in [(c, d)]$ implies that

$$c_1d = d_1c.$$

To get a "common denominator" (common second member) for the four pairs (a, b), (a_1, b_1), (c, d), and (c_1, d_1), we multiply the first equation by d_1d and the second equation by b_1b. Adding the resulting equations, we obtain the following equation in D:

$$a_1bd_1d + c_1db_1b = b_1ad_1d + d_1cb_1b.$$

Using various axioms for an integral domain, we see that

$$(a_1d_1 + b_1c_1)bd = b_1d_1(ad + bc),$$

so

$$(a_1d_1 + b_1c_1, b_1d_1) \sim (ad + bc, bd),$$

giving $(a_1d_1 + b_1c_1, b_1d_1) \in [(ad + bc, bd)]$. This takes care of addition in F. For multiplication in F, on multiplying the equations $a_1b = b_1a$ and $c_1d = d_1c$, we obtain

$$a_1bc_1d = b_1ad_1c,$$

so, using axioms of D, we get

$$a_1c_1bd = b_1d_1ac,$$

which implies that

$$(a_1c_1, b_1d_1) \sim (ac, bd).$$

Thus $(a_1c_1, b_1d_1) \in [(ac, bd)]$, which completes the proof. ◆

It is important to *understand* the meaning of the last lemma and the necessity for proving it. This completes our Step 2.

Step 3 Step 3 is routine, but it is good for us to work through a few of these details. The reason for this is that we cannot work through them unless we *understand* what we have done. Thus working through them will contribute to our understanding of this construction. We list the things that must be proved and prove a couple of them. The rest are left to the exercises.

1. Addition in F is commutative.

Proof Now $[(a, b)] + [(c, d)]$ is by definition $[(ad + bc, bd)]$. Also $[(c, d)] + [(a, b)]$ is by definition $[(cb + da, db)]$. We need to show that $(ad + bc, bd) \sim (cb + da, db)$. This is true, since $ad + bc = cb + da$ and $bd = db$, by the axioms of D. ◆

2. Addition is associative.

3. $[(0, 1)]$ is an identity element for addition in F.

4. $[(-a, b)]$ is an additive inverse for $[(a, b)]$ in F.

5. Multiplication in F is associative.

6. Multiplication in F is commutative.

7. The distributive laws hold in F.

8. $[(1, 1)]$ is a multiplicative identity element in F.

9. If $[(a, b)] \in F$ is not the additive identity element, then $a \neq 0$ in D and $[(b, a)]$ is a multiplicative inverse for $[(a, b)]$.

Proof Let $[(a, b)] \in F$. If $a = 0$, then

$$a1 = b0 = 0,$$

so

$$(a, b) \sim (0, 1),$$

that is, $[(a, b)] = [(0, 1)]$. But $[(0, 1)]$ is the additive identity by Part 3. Thus if $[(a, b)]$ is not the additive identity in F, we have $a \neq 0$, so it makes sense to talk about $[(b, a)]$ in F. Now $[(a, b)][(b, a)] = [(ab, ba)]$. But in D we have $ab = ba$, so $(ab)1 = (ba)1$, and

$$(ab, ba) \sim (1, 1).$$

Thus

$$[(a, b)][(b, a)] = [(1, 1)],$$

and $[(1, 1)]$ is the multiplicative identity by Part 8. ◆

This completes Step 3.

Step 4 It remains for us to show that F can be regarded as containing D. To do this, we show that there is an isomorphism i of D with a subdomain of F. Then if we rename the image of D under i using the names of the elements of D, we will be done. The next lemma gives us this isomorphism. We use the letter i for this isomorphism to suggest *injection* (see the footnote on page 4); we will inject D into F.

21.4 Lemma The map $i : D \to F$ given by $i(a) = [(a, 1)]$ is an isomorphism of D with a subring of F.

Proof For a and b in D, we have

$$i(a + b) = [(a + b, 1)].$$

Also,

$$i(a) + i(b) = [(a, 1)] + [(b, 1)] = [(a1 + 1b, 1)] = [(a + b, 1)].$$

so $i(a + b) = i(a) + i(b)$. Furthermore,

$$i(ab) = [(ab, 1)],$$

while

$$i(a)i(b) = [(a, 1)][(b, 1)] = [(ab, 1)],$$

so $i(ab) = i(a)i(b)$.

It remains for us to show only that i is one to one. If $i(a) = i(b)$, then

$$[(a, 1)] = [(b, 1)],$$

so $(a, 1) \sim (b, 1)$ giving $a1 = 1b$; that is,

$$a = b.$$

Thus i is an isomorphism of D with $i[D]$, and, of course, $i[D]$ is then a subdomain of F. ◆

Since $[(a, b)] = [(a, 1)][(1, b)] = [(a, 1)]/[(b, 1)] = i(a)/i(b)$ clearly holds in F, we have now proved the following theorem.

21.5 Theorem Any integral domain D can be enlarged to (or embedded in) a field F such that every element of F can be expressed as a quotient of two elements of D. (Such a field F is a **field of quotients of** D.)

Uniqueness

We said in the beginning that F could be regarded in some sense as a minimal field containing D. This is intuitively evident, since every field containing D must contain all elements a/b for every $a, b \in D$ with $b \neq 0$. The next theorem will show that every field containing D contains a subfield which is a field of quotients of D, and that any two fields of quotients of D are isomorphic.

21.6 Theorem Let F be a field of quotients of D and let L be any field containing D. Then there exists a map $\psi : F \to L$ that gives an isomorphism of F with a subfield of L such that $\psi(a) = a$ for $a \in D$.

Proof The subring and mapping diagram in Fig. 21.7 may help you to visualize the situation for this theorem.

An element of F is of the form $a /_F b$ where $/_F$ denotes the quotient of $a \in D$ by $b \in D$ regarded as elements of F. We of course want to map $a /_F b$ onto $a /_L b$ where $/_L$ denotes the quotient of elements in L. The main job will be to show that such a map is well defined.

We must define $\psi : F \to L$, and we start by defining

$$\psi(a) = a \qquad \text{for} \qquad a \in D.$$

Every $x \in F$ is a quotient $a /_F b$ of some two elements a and $b, b \neq 0$, of D. Let us attempt to define ψ by

$$\psi(a /_F b) = \psi(a) /_L \psi(b).$$

We must first show that this map ψ is sensible and well-defined. Since ψ is the identity on D, for $b \neq 0$ we have $\psi(b) \neq 0$, so our definition of $\psi(a /_F b)$ as $\psi(a) /_L \psi(b)$ makes sense. If $a /_F b = c /_F d$ in F, then $ad = bc$ in D, so $\psi(ad) = \psi(bc)$. But since ψ is the identity on D,

$$\psi(ad) = \psi(a)\psi(d) \qquad \text{and} \qquad \psi(bc) = \psi(b)\psi(c).$$

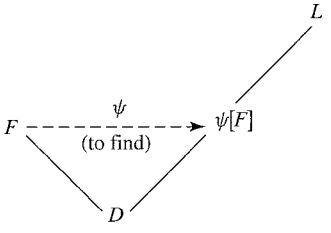
Thus

$$\psi(a) /_L \psi(b) = \psi(c) /_L \psi(d)$$

in L, so ψ is well-defined.

The equations

$$\psi(xy) = \psi(x)\psi(y)$$



21.7 Figure

and

$$\psi(x + y) = \psi(x) + \psi(y)$$

follow easily from the definition of ψ on F and from the fact that ψ is the identity on D.

If $\psi(a /_F b) = \psi(c /_F d)$, we have

$$\psi(a) /_L \psi(b) = \psi(c) /_L \psi(d)$$

so

$$\psi(a)\psi(d) = \psi(b)\psi(c).$$

Since ψ is the identity on D, we then deduce that $ad = bc$, so $a /_F b = c /_F d$. Thus ψ is one to one.

By definition, $\psi(a) = a$ for $a \in D$. ◆

21.8 Corollary Every field L containing an integral domain D contains a field of quotients of D.

Proof In the proof of Theorem 21.6 every element of the subfield $\psi[F]$ of L is a quotient in L of elements of D. ◆

21.9 Corollary Any two fields of quotients of an integral domain D are isomorphic.

Proof Suppose in Theorem 21.6 that L is a field of quotients of D, so that every element x of L can be expressed in the form $a /_L b$ for $a, b \in D$. Then L is the field $\psi[F]$ of the proof of Theorem 21.6 and is thus isomorphic to F. ◆

EXERCISES 21

Computations

1. Describe the field F of quotients of the integral subdomain

$$D = \{n + mi \mid n, m \in \mathbb{Z}\}$$

of $\mathbb{C}$. "Describe" means give the elements of $\mathbb{C}$ that make up the field of quotients of D in $\mathbb{C}$. (The elements of D are the **Gaussian integers.**)

2. Describe (in the sense of Exercise 1) the field F of quotients of the integral subdomain $D = \{n + m\sqrt{2} \mid n, m \in \mathbb{Z}\}$ of $\mathbb{R}$.

Concepts

3. Correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a form acceptable for publication.

A *field of quotients* of an integral domain D is a field F in which D can be embedded so that every nonzero element of D is a unit in F.

4. Mark each of the following true or false.

_____ **a.** $\mathbb{Q}$ is a field of quotients of $\mathbb{Z}$.

_____ **b.** $\mathbb{R}$ is a field of quotients of $\mathbb{Z}$.

_____ **c.** $\mathbb{R}$ is a field of quotients of $\mathbb{R}$.

_____ **d.** $\mathbb{C}$ is a field of quotients of $\mathbb{R}$.

_____ **e.** If D is a field, then any field of quotients of D is isomorphic to D.

_____ **f.** The fact that D has no divisors of 0 was used strongly several times in the construction of a field F of quotients of the integral domain D.

_____ **g.** Every element of an integral domain D is a unit in a field F of quotients of D.

_____ **h.** Every nonzero element of an integral domain D is a unit in a field F of quotients of D.

_____ **i.** A field of quotients F' of a subdomain D' of an integral domain D can be regarded as a subfield of some field of quotients of D.

_____ **j.** Every field of quotients of $\mathbb{Z}$ is isomorphic to $\mathbb{Q}$.

5. Show by an example that a field F' of quotients of a proper subdomain D' of an integral domain D may also be a field of quotients for D.

Theory

6. Prove Part 2 of Step 3. You may assume any preceding part of Step 3.

7. Prove Part 3 of Step 3. You may assume any preceding part of Step 3.

8. Prove Part 4 of Step 3. You may assume any preceding part of Step 3.

9. Prove Part 5 of Step 3. You may assume any preceding part of Step 3.

10. Prove Part 6 of Step 3. You may assume any preceding part of Step 3.

11. Prove Part 7 of Step 3. You may assume any preceding part of Step 3.

12. Let R be a nonzero commutative ring, and let T be a nonempty subset of R closed under multiplication and containing neither 0 nor divisors of 0. Starting with $R \times T$ and otherwise exactly following the construction in this section, we can show that the ring R can be enlarged to a *partial ring of quotients* $Q(R, T)$. Think about this for 15 minutes or so; look back over the construction and see why things still work. In particular, show the following:

a. $Q(R, T)$ has unity even if R does not.

b. In $Q(R, T)$, every nonzero element of T is a unit.

13. Prove from Exercise 12 that every nonzero commutative ring containing an element a that is not a divisor of 0 can be enlarged to a commutative ring with unity. Compare with Exercise 30 of Section 19.

14. With reference to Exercise 12, how many elements are there in the ring $Q(\mathbb{Z}_4, \{1, 3\})$?

15. With reference to Exercise 12, describe the ring $Q(\mathbb{Z}, \{2^n \mid n \in \mathbb{Z}^+\})$, by describing a subring of $\mathbb{R}$ to which it is isomorphic.

16. With reference to Exercise 12, describe the ring $Q(3\mathbb{Z}, \{6^n \mid n \in \mathbb{Z}^+\})$ by describing a subring of $\mathbb{R}$ to which it is isomorphic.

17. With reference to Exercise 12, suppose we drop the condition that T have no divisors of zero and just require that nonempty T not containing 0 be closed under multiplication. The attempt to enlarge R to a commutative ring with unity in which every nonzero element of T is a unit must fail if T contains an element a that is a divisor of 0, for a divisor of 0 cannot also be a unit. Try to discover where a construction parallel to that in the text but starting with $R \times T$ first runs into trouble. In particular, for $R = \mathbb{Z}_6$ and $T = \{1, 2, 4\}$, illustrate the first difficulty encountered. [*Hint:* It is in Step 1.]

SECTION 22 RINGS OF POLYNOMIALS

Polynomials in an Indeterminate

We all have a pretty workable idea of what constitutes a *polynomial in x with coefficients in a ring R*. We can guess how to add and multiply such polynomials and know what is meant by the *degree* of a polynomial. We expect that the set $R[x]$ of all polynomials with coefficients in the ring R is itself a ring with the usual operations of polynomial addition and multiplication, and that R is a subring of $R[x]$. However, we will be working with polynomials from a slightly different viewpoint than the approach in high school algebra or calculus, and there are a few things that we want to say.

In the first place, we will call x an **indeterminate** rather than a variable. Suppose, for example that our ring of coefficients is $\mathbb{Z}$. One of the polynomials in the ring $\mathbb{Z}[x]$ is $1x$, which we shall write simply as x. Now x is not 1 or 2 or any of the other elements of $\mathbb{Z}[x]$. Thus from now on we will never write such things as "$x = 1$" or "$x = 2$," as we have done in other courses. We call x an indeterminate rather than a variable to emphasize this change. Also, we will never write an expression such as "$x^2 - 4 = 0$," simply because $x^2 - 4$ is not the zero polynomial in our ring $\mathbb{Z}[x]$. We are accustomed to speaking of "solving a polynomial equation," and will be spending a lot of time in the remainder of our text discussing this, but we will always refer to it as "finding a zero of a polynomial." In summary, we try to be careful in our discussion of algebraic structures not to say in one context that things are equal and in another context that they are not equal.

HISTORICAL NOTE

The use of x and other letters near the end of the alphabet to represent an "indeterminate" is due to René Descartes (1596–1650). Earlier, François Viete (1540–1603) had used vowels for indeterminates and consonants for known quantities. Descartes is also responsible for the first publication of the factor theorem (Corollary 23.3) in his work *The Geometry,* which appeared as an appendix to his *Discourse on Method* (1637). This work also contained the first publication of the basic concepts of analytic geometry; Descartes showed how geometric curves can be described algebraically.

Descartes was born to a wealthy family in La Haye, France; since he was always of delicate health, he formed the habit of spending his mornings in bed. It was at these times that he accomplished his most productive work. The *Discourse on Method* was Descartes' attempt to show the proper procedures for "searching for truth in the sciences." The first step in this process was to reject as absolutely false everything of which he had the least doubt; but, since it was necessary that he who was thinking was "something," he conceived his first principle of philosophy: "I think, therefore I am." The most enlightening parts of the *Discourse on Method,* however, are the three appendices: *The Optics, The Geometry,* and *The Meteorology*. It was here that Descartes provided examples of how he actually applied his method. Among the important ideas Descartes discovered and published in these works were the sine law of refraction of light, the basics of the theory of equations, and a geometric explanation of the rainbow.

In 1649, Descartes was invited by Queen Christina of Sweden to come to Stockholm to tutor her. Unfortunately, the Queen required him, contrary to his long-established habits, to rise at an early hour. He soon contracted a lung disease and died in 1650.

If a person knows nothing about polynomials, it is not an easy task to describe precisely the nature of a polynomial in x with coefficients in a ring R. If we just define such a polynomial to be a *finite formal sum*

$$\sum_{i=0}^{n} a_i x^i = a_0 + a_1 x + \cdots + a_n x^n,$$

where $a_i \in R$, we get ourselves into a bit of trouble. For surely $0 + a_1 x$ and $0 + a_1 x + 0x^2$ are different as formal sums, but we want to regard them as the same polynomial. A practical solution to this problem is to define a polynomial as an *infinite formal sum*

$$\sum_{i=0}^{\infty} a_i x^i = a_0 + a_1 x + \cdots + a_n x^n + \cdots,$$

where $a_i = 0$ for all but a finite number of values of i. Now there is no problem of having more than one formal sum represent what we wish to consider a single polynomial.

22.1 Definition Let R be a ring. A **polynomial** $f(x)$ **with coefficients in** R is an infinite formal sum

$$\sum_{i=0}^{\infty} a_i x^i = a_0 + a_1 x + \cdots + a_n x^n + \cdots,$$

where $a_i \in R$ and $a_i = 0$ for all but a finite number of values of i. The a_i are **coefficients of** $f(x)$. If for some $i \geq 0$ it is true that $a_i \neq 0$, the largest such value of i is the **degree of** $f(x)$. If all $a_i = 0$, then the degree of $f(x)$ is undefined.[†] ■

To simplify working with polynomials, let us agree that if $f(x) = a_0 + a_1 x + \cdots + a_n x^n + \cdots$ has $a_i = 0$ for $i > n$, then we may denote $f(x)$ by $a_0 + a_1 x + \cdots + a_n x^n$. Also, if R has unity $1 \neq 0$, we will write a term $1x^k$ in such a sum as x^k. For example, in $\mathbb{Z}[x]$, we will write the polynomial $2 + 1x$ as $2 + x$. Finally, we shall agree that we may omit altogether from the formal sum any term $0x^i$, or a_0 if $a_0 = 0$ but not all $a_i = 0$. Thus 0, 2, x, and $2 + x^2$ are polynomials with coefficients in $\mathbb{Z}$. An element of R is a **constant polynomial.**

Addition and multiplication of polynomials with coefficients in a ring R are defined in a way familiar to us. If

$$f(x) = a_0 + a_1 x + \cdots + a_n x^n + \cdots$$

and

$$g(x) = b_0 + b_1 x + \cdots + b_n x^n + \cdots,$$

then for polynomial addition, we have

$$f(x) + g(x) = c_0 + c_1 x + \cdots + c_n x^n + \cdots \text{ where } c_n = a_n + b_n,$$

[†] The degree of the zero polynomial is sometimes defined to be -1, which is the first integer less than 0, or defined to be $-\infty$ so that the degree of $f(x)g(x)$ will be the sum of the degrees of $f(x)$ and $g(x)$ if one of them is zero.

and for polynomial multiplication, we have

$$f(x)g(x) = d_0 + d_1x + \cdots + d_nx^n + \cdots \text{ where } d_n = \sum_{i=0}^{n} a_i b_{n-i}$$

Observe that both c_i and d_i are 0 for all but a finite number of values of i, so these definitions make sense. Note that $\sum_{i=0}^{n} a_i b_{n-i}$ need not equal $\sum_{i=0}^{n} b_i a_{n-i}$ if R is not commutative. With these definitions of addition and multiplication, we have the following theorem.

22.2 Theorem The set $R[x]$ of all polynomials in an indeterminate x with coefficients in a ring R is a ring under polynomial addition and multiplication. If R is commutative, then so is $R[x]$, and if R has unity $1 \neq 0$, then 1 is also unity for $R[x]$.

Proof That $\langle R[x], + \rangle$ is an abelian group is apparent. The associative law for multiplication and the distributive laws are straightforward, but slightly cumbersome, computations. We illustrate by proving the associative law.

Applying ring axioms to $a_i, b_j, c_k \in R$, we obtain

$$\begin{aligned}
\left[\left(\sum_{i=0}^{\infty} a_i x^i\right)\left(\sum_{j=0}^{\infty} b_j x^j\right)\right]\left(\sum_{k=0}^{\infty} c_k x^k\right) &= \left[\sum_{n=0}^{\infty}\left(\sum_{i=0}^{n} a_i b_{n-i}\right)x^n\right]\left(\sum_{k=0}^{\infty} c_k x^k\right) \\
&= \sum_{s=0}^{\infty}\left[\sum_{n=0}^{s}\left(\sum_{i=0}^{n} a_i b_{n-i}\right)c_{s-n}\right]x^s \\
&= \sum_{s=0}^{\infty}\left(\sum_{i+j+k=s} a_i b_j c_k\right)x^s \\
&= \sum_{s=0}^{\infty}\left[\sum_{m=0}^{s} a_{s-m}\left(\sum_{j=0}^{m} b_j c_{m-j}\right)\right]x^s \\
&= \left(\sum_{i=0}^{\infty} a_i x^i\right)\left[\sum_{m=0}^{\infty}\left(\sum_{j=0}^{m} b_j c_{m-j}\right)x^m\right] \\
&= \left(\sum_{i=0}^{\infty} a_i x^i\right)\left[\left(\sum_{j=0}^{\infty} b_j x^j\right)\left(\sum_{k=0}^{\infty} c_k x^k\right)\right].
\end{aligned}$$

Whew!! In this computation, the fourth expression, having just two summation signs, should be viewed as the value of the triple product $f(x)g(x)h(x)$ of these polynomials under this associative multiplication. (In a similar fashion, we view $f(g(h(x)))$ as the value of the associative composition $(f \circ g \circ h)(x)$ of three functions f, g, and h.)

The distributive laws are similarly proved. (See Exercise 26.)

The comments prior to the statement of the theorem show that $R[x]$ is a commutative ring if R is commutative, and a unity $1 \neq 0$ in R is also unity for $R[x]$, in view of the definition of multiplication in $R[x]$. ◆

Thus $\mathbb{Z}[x]$ is the ring of polynomials in the indeterminate x with integral coefficients, $Q[x]$ the ring of polynomials in x with rational coefficients, and so on.

22.3 Example In $\mathbb{Z}_2[x]$, we have

$$(x+1)^2 = (x+1)(x+1) = x^2 + (1+1)x + 1 = x^2 + 1.$$

Still working in $\mathbb{Z}_2[x]$, we obtain

$$(x+1) + (x+1) = (1+1)x + (1+1) = 0x + 0 = 0. \quad \blacktriangle$$

If R is a ring and x and y are two indeterminates, then we can form the ring $(R[x])[y]$, that is, the ring of polynomials in y with coefficients that are polynomials in x. Every polynomial in y with coefficients that are polynomials in x can be rewritten in a natural way as a polynomial in x with coefficients that are polynomials in y as illustrated by Exercise 20. This indicates that $(R[x])[y]$ is naturally isomorphic to $(R[y])[x]$, although a careful proof is tedious. We shall identify these rings by means of this natural isomorphism, and shall consider this ring $R[x, y]$ the **ring of polynomials in two indeterminates x and y with coefficients in R**. The **ring $R[x_1, \cdots, x_n]$ of polynomials in the n indeterminates x_i with coefficients in R** is similarly defined.

We leave as Exercise 24 the proof that if D is an integral domain then so is $D[x]$. In particular, if F is a field, then $F[x]$ is an integral domain. Note that $F[x]$ is not a field, for x is not a unit in $F[x]$. That is, there is no polynomial $f(x) \in F[x]$ such that $xf(x) = 1$. By Theorem 21.5, one can construct the field of quotients $F(x)$ of $F[x]$. Any element in $F(x)$ can be represented as a quotient $f(x)/g(x)$ of two polynomials in $F[x]$ with $g(x) \neq 0$. We similarly define $F(x_1, \cdots, x_n)$ to be the field of quotients of $F[x_1, \cdots, x_n]$. This field $F(x_1, \cdots, x_n)$ is the **field of rational functions in n indeterminates over F**. These fields play a very important role in algebraic geometry.

The Evaluation Homomorphisms

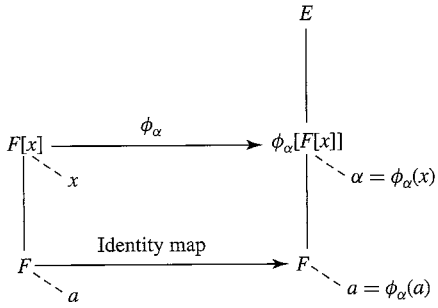
We are now ready to proceed to show how homomorphisms can be used to study what we have always referred to as "solving a polynomial equation." Let E and F be fields, with F a subfield of E, that is, $F \leq E$. The next theorem asserts the existence of very important homomorphisms of $F[x]$ into E. *These homomorphisms will be the fundamental tools for much of the rest of our work.*

22.4 Theorem **(The Evaluation Homomorphisms for Field Theory)** Let F be a subfield of a field E, let α be any element of E, and let x be an indeterminate. The map $\phi_\alpha : F[x] \to E$ defined by

$$\phi_\alpha(a_0 + a_1x + \cdots + a_nx^n) = a_0 + a_1\alpha + \cdots + a_n\alpha^n$$

for $(a_0 + a_1x + \cdots + a_nx^n) \in F[x]$ is a homomorphism of $F[x]$ into E. Also, $\phi_\alpha(x) = \alpha$, and ϕ_α maps F isomorphically by the identity map; that is, $\phi_\alpha(a) = a$ for $a \in F$. The homomorphism ϕ_α is **evaluation at α**.

Proof The subfield and mapping diagram in Fig. 22.5 may help us to visualize this situation. The dashed lines indicate an element of the set. The theorem is really an immediate



22.5 Figure

consequence of our definitions of addition and multiplication in $F[x]$. The map ϕ_α is well defined, that is, independent of our representation of $f(x) \in F[x]$ as a finite sum

$$a_0 + a_1x + \cdots + a_nx^n,$$

since such a finite sum representing $f(x)$ can be changed only by insertion or deletion of terms $0x^i$, which does not affect the value of $\phi_\alpha(f(x))$.

If $f(x) = a_0 + a_1x + \cdots + a_nx^n$, $g(x) = b_0 + b_1x + \cdots + b_mx^m$, and $h(x) = f(x) + g(x) = c_0 + c_1x + \cdots + c_rx^r$, then

$$\phi_\alpha(f(x) + g(x)) = \phi_\alpha(h(x)) = c_0 + c_1\alpha + \cdots + c_r\alpha^r,$$

while

$$\phi_\alpha(f(x)) + \phi_\alpha(g(x)) = \big(a_0 + a_1\alpha + \cdots + a_n\alpha^n\big) + \big(b_0 + b_1\alpha + \cdots + b_m\alpha^m\big).$$

Since by definition of polynomial addition we have $c_i = a_i + b_i$, we see that

$$\phi_\alpha(f(x) + g(x)) = \phi_\alpha(f(x)) + \phi_\alpha(g(x)).$$

Turning to multiplication, we see that if

$$f(x)g(x) = d_0 + d_1x + \cdots + d_sx^s,$$

then

$$\phi_\alpha(f(x)g(x)) = d_0 + d_1\alpha + \cdots + d_s\alpha^s,$$

while

$$[\phi_\alpha(f(x))][\phi_\alpha(g(x))] = \big(a_0 + a_1\alpha + \cdots + \alpha_n\alpha^n\big)\big(b_0 + b_1\alpha + \cdots + b_m\alpha^m\big).$$

Since by definition of polynomial multiplication $d_j = \sum_{i=0}^{j} a_i b_{j-i}$, we see that

$$\phi_\alpha(f(x)g(x)) = [\phi_\alpha(f(x))][\phi_\alpha(g(x))].$$

Thus ϕ_α is a homomorphism.

The very definition of ϕ_α applied to a constant polynomial $a \in F[x]$, where $a \in F$, gives $\phi_\alpha(a) = a$, so ϕ_α maps F isomorphically by the identity map. Again by definition of ϕ_α, we have $\phi_\alpha(x) = \phi_\alpha(1x) = 1\alpha = \alpha$. ◆

We point out that this theorem is valid with the identical proof if F and E are merely commutative rings with unity rather than fields. However, we shall be interested primarily in the case in which they are fields.

It is hard to overemphasize the importance of this simple theorem for us. It is the very foundation for all of our further work in field theory. It is so simple that it could justifiably be called an *observation* rather than a theorem. It was perhaps a little misleading to write out the proof because the polynomial notation makes it look so complicated that you may be fooled into thinking it is a difficult theorem.

22.6 Example Let F be $\mathbb{Q}$ and E be $\mathbb{R}$ in Theorem 22.4, and consider the evaluation homomorphism $\phi_0 : \mathbb{Q}[x] \to \mathbb{R}$. Here

$$\phi_0(a_0 + a_1x + \cdots + a_nx^n) = a_0 + a_10 + \cdots + a_n0^n = a_0.$$

Thus every polynomial is mapped onto its constant term. ▲

22.7 Example Let F be $\mathbb{Q}$ and E be $\mathbb{R}$ in Theorem 22.4 and consider the evaluation homomorphism $\phi_2 : \mathbb{Q}[x] \to \mathbb{R}$. Here

$$\phi_2(a_0 + a_1x + \cdots + a_nx^n) = a_0 + a_12 + \cdots + a_n2^n.$$

Note that

$$\phi_2(x^2 + x - 6) = 2^2 + 2 - 6 = 0.$$

Thus $x^2 + x - 6$ is in the kernel N of ϕ_2. Of course,

$$x^2 + x - 6 = (x - 2)(x + 3),$$

and the reason that $\phi_2(x^2 + x - 6) = 0$ is that $\phi_2(x - 2) = 2 - 2 = 0$. ▲

22.8 Example Let F be $\mathbb{Q}$ and E be $\mathbb{C}$ in Theorem 22.4 and consider the evaluation homomorphism $\phi_i : \mathbb{Q}[x] \to \mathbb{C}$. Here

$$\phi_i(a_0 + a_1x + \cdots + a_nx^n) = a_0 + a_1i + \cdots + a_ni^n$$

and $\phi_i(x) = i$. Note that

$$\phi_i(x^2 + 1) = i^2 + 1 = 0,$$

so $x^2 + 1$ is in the kernel N of ϕ_i. ▲

22.9 Example Let F be $\mathbb{Q}$ and let E be $\mathbb{R}$ in Theorem 22.4 and consider the evaluation homomorphism $\phi_\pi : \mathbb{Q}[x] \to \mathbb{R}$. Here

$$\phi_\pi(a_0 + a_1x + \cdots + a_nx^n) = a_0 + a_1\pi + \cdots + a_n\pi^n.$$

It can be proved that $a_0 + a_1\pi + \cdots + a_n\pi^n = 0$ if and only if $a_i = 0$ for $i = 0, 1, \cdots, n$. Thus the kernel of ϕ_π is $\{0\}$, and ϕ_π is a one-to-one map. This shows that all *formal polynomials in π with rational coefficients* form a ring isomorphic to $\mathbb{Q}[x]$ in a natural way with $\phi_\pi(x) = \pi$. ▲

The New Approach

We now complete the connection between our new ideas and the classical concept of solving a polynomial equation. Rather than speak of *solving a polynomial equation,* we shall refer to *finding a zero of a polynomial.*

22.10 Definition Let F be a subfield of a field E, and let α be an element of E. Let $f(x) = a_0 + a_1x + \cdots + a_nx^n$ be in $F[x]$, and let $\phi_\alpha : F[x] \to E$ be the evaluation homomorphism of Theorem 22.4. Let $f(\alpha)$ denote

$$\phi_\alpha(f(x)) = a_0 + a_1\alpha + \cdots + a_n\alpha^n.$$

If $f(\alpha) = 0$, then α is a **zero of** $f(x)$. ■

In terms of this definition, we can rephrase the classical problem of finding all real numbers r such that $r^2 + r - 6 = 0$ by letting $F = \mathbb{Q}$ and $E = \mathbb{R}$ and *finding all $\alpha \in \mathbb{R}$ such that*

$$\phi_\alpha(x^2 + x - 6) = 0,$$

that is, finding all zeros of $x^2 + x - 6$ in $\mathbb{R}$. Both problems have the same answer, since

$$\{\alpha \in \mathbb{R} \mid \phi_\alpha(x^2 + x - 6) = 0\} = \{r \in \mathbb{R} \mid r^2 + r - 6 = 0\} = \{2, -3\}.$$

It may seem that we have merely succeeded in making a simple problem seem quite complicated. In fact, *what we have done is to phrase the problem in the language of mappings, and we can now use all the mapping machinery that we have developed and will continue to develop for its solution.*

Our Basic Goal

We continue to attempt to put our future work in perspective. Sections 26 and 27 are concerned with topics in ring theory that are analogous to the material on factor groups and homomorphisms for group theory. However, our aim in developing these analogous concepts for rings will be quite different from our aims in group theory. In group theory we used the concepts of factor groups and homomorphisms to study the structure of a given group and to determine the types of group structures of certain orders that could exist. We will be talking about homomorphisms and factor rings in Section 26

with an eye to finding zeros of polynomials, which is one of the oldest and most fundamental problems in algebra. Let us take a moment to talk about this aim in the light of mathematical history, using the language of "solving polynomial equations" to which we are accustomed.

We start with the Pythagorean school of mathematics of about 525 B.C. The Pythagoreans worked with the assumption that all distances are **commensurable**: that is, given distances a and b, there should exist a unit of distance u and integers n and m such that $a = (n)(u)$ and $b = (m)(u)$. In terms of numbers, then, thinking of u as being one unit of distance, they maintained that all numbers are integers. This idea of commensurability can be rephrased according to our ideas as an assertion that all numbers are rational, for if a and b are rational numbers, then each is an integral multiple of the reciprocal of the least common multiple of their denominators. For example, if $a = \frac{7}{12}$ and $b = \frac{19}{15}$, then $a = (35)(\frac{1}{60})$ and $b = (76)(\frac{1}{60})$.

The Pythagoreans knew, of course, what is now called the *Pythagorean theorem;* that is, for a right triangle with legs of lengths a and b and a hypotenuse of length c,

$$a^2 + b^2 = c^2.$$

They also had to grant the existence of a hypotenuse of a right triangle having two legs of equal length, say one unit each. The hypotenuse of such a right triangle would, as we know, have to have a length of $\sqrt{2}$. Imagine then their consternation and dismay when one of their society—according to some stories it was Pythagoras himself—came up with the embarrassing fact that is stated in our terminology in the following theorem.

22.11 Theorem The polynomial $x^2 - 2$ has no zeros in the rational numbers. Thus $\sqrt{2}$ is not a rational number.

Proof Suppose that m/n for $m, n \in \mathbb{Z}$ is a rational number such that $(m/n)^2 = 2$. We assume that we have canceled any factors common to m and n, so that the fraction m/n is in lowest terms with $\gcd(m, n) = 1$. Then

$$m^2 = 2n^2,$$

where both m^2 and $2n^2$ are integers. Since m^2 and $2n^2$ are the same integer, and since 2 is a factor of $2n^2$, we see that 2 must be one of the factors of m^2. But as a square, m^2 has as factors the factors of m repeated twice. Thus m^2 must have two factors 2. Then $2n^2$ must have two factors 2, so n^2 must have 2 as a factor, and consequently n has 2 as a factor. We have deduced from $m^2 = 2n^2$ that both m and n must be divisible by 2, contradicting the fact that the fraction m/n is in lowest terms. Thus we have $2 \neq (m/n)^2$ for any $m, n \in \mathbb{Z}$. ◆

Thus the Pythagoreans ran right into the question of a solution of a polynomial equation, $x^2 - 2 = 0$. We refer the student to Shanks [36, Chapter 3], for a lively and totally delightful account of this Pythagorean dilemma and its significance in mathematics.

Historical Note

The solution of polynomial equations has been a goal of mathematics for nearly 4000 years. The Babylonians developed versions of the quadratic formula to solve quadratic equations. For example, to solve $x^2 - x = 870$, the Babylonian scribe instructed his students to take half of 1 ($\frac{1}{2}$), square it ($\frac{1}{4}$), and add that to 870. The square root of $870\frac{1}{4}$, namely $29\frac{1}{2}$, is then added to $\frac{1}{2}$ to give 30 as the answer. What the scribes did not discuss, however, was what to do if the square root in this process was not a rational number. Chinese mathematicians, however, from about 200 B.C., discovered a method similar to what is now called *Horner's method* to solve quadratic equations numerically; since they used a decimal system, they were able in principle to carry out the computation to as many places as necessary and could therefore ignore the distinction between rational and irrational solutions. The Chinese, in fact, extended their numerical techniques to polynomial equations of higher degree. In the Arab world, the Persian poet–mathematician Omar Khayyam (1048–1131) developed methods for solving cubic equations geometrically by finding the point(s) of intersection of appropriately chosen conic sections, while Sharaf al-Din al-Tusi (died 1213) used, in effect, techniques of calculus to determine whether or not a cubic equation had a real positive root. It was the Italian Girolamo Cardano (1501–1576) who first published a procedure for solving cubic equations algebraically.

In our motivation of the definition of a group, we commented on the necessity of having negative numbers, so that equations such as $x + 2 = 0$ might have solutions. The introduction of negative numbers caused a certain amount of consternation in some philosophical circles. We can visualize 1 apple, 2 apples, and even $\frac{13}{11}$ apples, but how can we point to anything and say that it is -17 apples? Finally, consideration of the equation $x^2 + 1 = 0$ led to the introduction of the number i. The very name of an "imaginary number" given to i shows how this number was regarded. Even today, many students are led by this name to regard i with some degree of suspicion. The negative numbers were introduced to us at such an early stage in our mathematical development that we accepted them without question.

We first met polynomials in high school freshman algebra. The first problem there was to learn how to add, multiply, and factor polynomials. Then, in both freshman algebra and in the second course in algebra in high school, considerable emphasis was placed on solving polynomial equations. These topics are exactly those with which we shall be concerned. The difference is that while in high school, only polynomials with real number coefficients were considered, *we shall be doing our work for polynomials with coefficients from any field.*

Once we have developed the machinery of homomorphisms and factor rings in Section 26, we will proceed with our **basic goal**: to show that given any polynomial of degree ≥ 1, where the coefficients of the polynomial may be from any field, we can find a zero of this polynomial in some field containing the given field. After the machinery is developed in Sections 26 and 27, the achievement of this goal will be very easy, and is really a very elegant piece of mathematics.

All this fuss may seem ridiculous, but just think back in history. This is the *culmination of more than 2000 years of mathematical endeavor in working with polynomial equations.* After achieving our *basic goal,* we shall spend the rest of our time studying the

nature of these solutions of polynomial equations. We need have no fear in approaching this material. *We shall be dealing with familiar topics of high school algebra. This work should seem much more natural than group theory.*

In conclusion, we remark that the machinery of factor rings and ring homomorphisms is not really necessary in order for us to achieve our *basic goal*. For a direct demonstration, see Artin [27, p. 29]. However, factor rings and ring homomorphisms are fundamental ideas that we should grasp, and our *basic goal* will follow very easily once we have mastered them.

EXERCISES 22

Computations

In Exercises 1 through 4, find the sum and the product of the given polynomials in the given polynomial ring.

1. $f(x) = 4x - 5$, $g(x) = 2x^2 - 4x + 2$ in $\mathbb{Z}_8[x]$.
2. $f(x) = x + 1$, $g(x) = x + 1$ in $\mathbb{Z}_2[x]$.
3. $f(x) = 2x^2 + 3x + 4$, $g(x) = 3x^2 + 2x + 3$ in $\mathbb{Z}_6[x]$.
4. $f(x) = 2x^3 + 4x^2 + 3x + 2$, $g(x) = 3x^4 + 2x + 4$ in $\mathbb{Z}_5[x]$.
5. How many polynomials are there of degree ≤ 3 in $\mathbb{Z}_2[x]$? (Include 0.)
6. How many polynomials are there of degree ≤ 2 in $\mathbb{Z}_5[x]$? (Include 0.)

In Exercises 7 and 8, $F = E = \mathbb{C}$ in Theorem 22.4. Compute for the indicated evaluation homomorphism.

7. $\phi_2(x^2 + 3)$
8. $\phi_i(2x^3 - x^2 + 3x + 2)$

In Exercises 9 through 11, $F = E = \mathbb{Z}_7$ in Theorem 22.4. Compute for the indicated evaluation homomorphism.

9. $\phi_3[(x^4 + 2x)(x^3 - 3x^2 + 3)]$
10. $\phi_5[(x^3 + 2)(4x^2 + 3)(x^7 + 3x^2 + 1)]$
11. $\phi_4(3x^{106} + 5x^{99} + 2x^{53})$ [*Hint:* Use Fermat's theorem.]

In Exercises 12 through 15, find all zeros in the indicated finite field of the given polynomial with coefficients in that field. [*Hint:* One way is simply to try all candidates!]

12. $x^2 + 1$ in $\mathbb{Z}_2$
13. $x^3 + 2x + 2$ in $\mathbb{Z}_7$
14. $x^5 + 3x^3 + x^2 + 2x$ in $\mathbb{Z}_5$
15. $f(x)g(x)$ where $f(x) = x^3 + 2x^2 + 5$ and $g(x) = 3x^2 + 2x$ in $\mathbb{Z}_7$
16. Let $\phi_a : \mathbb{Z}_5[x] \to \mathbb{Z}_5$ be an evaluation homomorphism as in Theorem 22.4. Use Fermat's theorem to evaluate $\phi_3(x^{231} + 3x^{117} - 2x^{53} + 1)$.
17. Use Fermat's theorem to find all zeros in $\mathbb{Z}_5$ of $2x^{219} + 3x^{74} + 2x^{57} + 3x^{44}$.

Concepts

In Exercises 18 and 19, correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a form acceptable for publication.

18. A *polynomial with coefficients in a ring R is an infinite formal sum*

$$\sum_{i=0}^{\infty} a_i x^i = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n + \cdots$$

where $a_i \in R$ for $i = 0, 1, 2, \cdots$.

19. Let F be a field and let $f(x) \in F[x]$. A *zero of* $f(x)$ is an $\alpha \in F$ such that $\phi_\alpha(f(x)) = 0$, where $\phi_\alpha : F(x) \to F$ is the evaluation homomorphism mapping x into α.

20. Consider the element

$$f(x, y) = (3x^3 + 2x)y^3 + (x^2 - 6x + 1)y^2 + (x^4 - 2x)y + (x^4 - 3x^2 + 2)$$

of $(\mathbb{Q}[x])[y]$. Write $f(x, y)$ as it would appear if viewed as an element of $(\mathbb{Q}[y])[x]$.

21. Consider the evaluation homomorphism $\phi_5 : \mathbb{Q}[x] \to \mathbb{R}$. Find six elements in the kernel of the homomorphism ϕ_5.

22. Find a polynomial of degree >0 in $\mathbb{Z}_4[x]$ that is a unit.

23. Mark each of the following true or false.

_____ **a.** The polynomial $(a_n x^n + \cdots + a_1 x + a_0) \in R[x]$ is 0 if and only if $a_i = 0$, for $i = 0, 1, \cdots, n$.

_____ **b.** If R is a commutative ring, then $R[x]$ is commutative.

_____ **c.** If D is an integral domain, then $D[x]$ is an integral domain.

_____ **d.** If R is a ring containing divisors of 0, then $R[x]$ has divisors of 0.

_____ **e.** If R is a ring and $f(x)$ and $g(x)$ in $R[x]$ are of degrees 3 and 4, respectively, then $f(x)g(x)$ may be of degree 8 in $R[x]$.

_____ **f.** If R is any ring and $f(x)$ and $g(x)$ in $R[x]$ are of degrees 3 and 4, respectively, then $f(x)g(x)$ is always of degree 7.

_____ **g.** If F is a subfield E and $\alpha \in E$ is a zero of $f(x) \in F[x]$, then α is a zero of $h(x) = f(x)g(x)$ for all $g(x) \in F[x]$.

_____ **h.** If F is a field, then the units in $F[x]$ are precisely the units in F.

_____ **i.** If R is a ring, then x is never a divisor of 0 in $R[x]$.

_____ **j.** If R is a ring, then the zero divisors in $R[x]$ are precisely the zero divisors in R.

Theory

24. Prove that if D is an integral domain, then $D[x]$ is an integral domain.

25. Let D be an integral domain and x an indeterminate.

a. Describe the units in $D[x]$.

b. Find the units in $\mathbb{Z}[x]$.

c. Find the units in $\mathbb{Z}_7[x]$.

26. Prove the left distributive law for $R[x]$, where R is a ring and x is an indeterminate.

27. Let F be a field of characteristic zero and let D be the formal polynomial differentiation map, so that

$$D(a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n) = a_1 + 2 \cdot a_2 x + \cdots + n \cdot a_n x^{n-1}.$$

a. Show that $D : F[x] \to F[x]$ is a group homomorphism of $\langle F[x], + \rangle$ into itself. Is D a ring homomorphism?

b. Find the kernel of D.

c. Find the image of $F[x]$ under D.

28. Let F be a subfield of a field E.

a. Define an *evaluation homomorphism*

$$\phi_{\alpha_1,\cdots,\alpha_n} : F[x_1, \cdots, x_n] \to E \qquad \text{for} \quad \alpha_i \in E,$$

stating the analog of Theorem 22.4.

b. With $E = F = \mathbb{Q}$, compute $\phi_{-3,2}({x_1}^2{x_2}^3 + 3{x_1}^4 x_2)$.

c. Define the concept of a *zero of a polynomial* $f(x_1, \cdots, x_n) \in F[x_1, \cdots, x_n]$ in a way analogous to the definition in the text of a zero of $f(x)$.

29. Let R be a ring, and let R^R be the set of all functions mapping R into R. For $\phi, \psi \in R^R$, define the sum $\phi + \psi$ by

$$(\phi + \psi)(r) = \phi(r) + \psi(r)$$

and the product $\phi \cdot \psi$ by

$$(\phi \cdot \psi)(r) = \phi(r)\psi(r)$$

for $r \in R$. Note that $\cdot$ is *not* function composition. Show that $\langle R^R, +, \cdot \rangle$ is a ring.

30. Referring to Exercise 29, let F be a field. An element ϕ of F^F is a **polynomial function on** F, if there exists $f(x) \in F[x]$ such that $\phi(a) = f(a)$ for all $a \in F$.

a. Show that the set P_F of all polynomial functions on F forms a subring of F^F.

b. Show that the ring P_F is not necessarily isomorphic to $F[x]$. [*Hint:* Show that if F is a finite field, P_F and $F[x]$ don't even have the same number of elements.]

31. Refer to Exercises 29 and 30 for the following questions.

a. How many elements are there in ${\mathbb{Z}_2}^{\mathbb{Z}_2}$? in ${\mathbb{Z}_3}^{\mathbb{Z}_3}$?

b. Classify $\langle {\mathbb{Z}_2}^{\mathbb{Z}_2}, + \rangle$ and $\langle {\mathbb{Z}_3}^{\mathbb{Z}_3}, + \rangle$ by Theorem 11.12, the Fundamental Theorem of finitely generated abelian groups.

c. Show that if F is a finite field, then $F^F = P_F$. [*Hint:* Of course, $P_F \subseteq F^F$. Let F have as elements $a_1, \cdots, a_n$. Note that if

$$f_i(x) = c(x - a_1) \cdots (x - a_{i-1})(x - a_{i+1}) \cdots (x - a_n),$$

then $f_i(a_j) = 0$ for $j \neq i$, and the value $f_i(a_i)$ can be controlled by the choice of $c \in F$. Use this to show that every function on F is a polynomial function.]

SECTION 23 Factorization of Polynomials over a Field

Recall that we are concerned with finding zeros of polynomials. Let E and F be fields, with $F \leq E$. Suppose that $f(x) \in F[x]$ factors in $F[x]$, so that $f(x) = g(x)h(x)$ for $g(x), h(x) \in F[x]$ and let $\alpha \in E$. Now for the evaluation homomorphism ϕ_α, we have

$$f(\alpha) = \phi_\alpha(f(x)) = \phi_\alpha(g(x)h(x)) = \phi_\alpha(g(x))\phi_\alpha(h(x)) = g(\alpha)h(\alpha).$$

Thus if $\alpha \in E$, then $f(\alpha) = 0$ if and only if either $g(\alpha) = 0$ or $h(\alpha) = 0$. The attempt to find a zero of $f(x)$ is reduced to the problem of finding a zero of a factor of $f(x)$. This is one reason why it is useful to study factorization of polynomials.

The Division Algorithm in $F[x]$

The following theorem is the basic tool for our work in this section. Note the similarity with the division algorithm for $\mathbb{Z}$ given in Theorem 6.3, the importance of which has been amply demonstrated.

23.1 Theorem **(Division Algorithm for $F[x]$)** Let

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$$

and

$$g(x) = b_m x^m + b_{m-1}x^{m-1} + \cdots + b_0$$

be two elements of $F[x]$, with a_n and b_m both nonzero elements of F and $m > 0$. Then there are unique polynomials $q(x)$ and $r(x)$ in $F[x]$ such that $f(x) = g(x)q(x) + r(x)$, where either $r(x) = 0$ or the degree of $r(x)$ is less than the degree m of $g(x)$.

Proof Consider the set $S = \{f(x) - g(x)s(x) \mid s(x) \in F[x]\}$. If $0 \in S$ then there exists an $s(x)$ such that $f(x) - g(x)s(x) = 0$, so $f(x) = g(x)s(x)$. Taking $q(x) = s(x)$ and $r(x) = 0$, we are done. Otherwise, let $r(x)$ be an element of minimal degree in S. Then

$$f(x) = g(x)q(x) + r(x)$$

for some $q(x) \in F[x]$. We must show that the degree of $r(x)$ is less than m. Suppose that

$$r(x) = c_t x^t + c_{t-1}x^{t-1} + \cdots + c_0,$$

with $c_j \in F$ and $c_t \neq 0$. If $t \geq m$, then

$$f(x) - q(x)g(x) - (c_t/b_m)x^{t-m}g(x) = r(x) - (c_t/b_m)x^{t-m}g(x), \tag{1}$$

and the latter is of the form

$$r(x) - (c_t x^t + \text{terms of lower degree}),$$

which is a polynomial of degree lower than t, the degree of $r(x)$. However, the polynomial in Eq. (1) can be written in the form

$$f(x) - g(x)\big[q(x) + (c_t/b_m)x^{t-m}\big],$$

so it is in S, contradicting the fact that $r(x)$ was selected to have minimal degree in S. Thus the degree of $r(x)$ is less than the degree m of $g(x)$.

For uniqueness, if

$$f(x) = g(x)q_1(x) + r_1(x)$$

and

$$f(x) = g(x)q_2(x) + r_2(x),$$

then subtracting we have

$$g(x)[q_1(x) - q_2(x)] = r_2(x) - r_1(x).$$

Because either $r_2(x) - r_1(x) = 0$ or the degree of $r_2(x) - r_1(x)$ is less than the degree of $g(x)$, this can hold only if $q_1(x) - q_2(x) = 0$ so $q_1(x) = q_2(x)$. Then we must have $r_2(x) - r_1(x) = 0$ so $r_1(x) = r_2(x)$. ◆

We can compute the polynomials $q(x)$ and $r(x)$ of Theorem 23.1 by long division just as we divided polynomials in $\mathbb{R}[x]$ in high school.

23.2 Example Let us work with polynomials in $\mathbb{Z}_5[x]$ and divide

$$f(x) = x^4 - 3x^3 + 2x^3 + 4x - 1$$

by $g(x) = x^2 - 2x + 3$ to find $q(x)$ and $r(x)$ of Theorem 23.1. The long division should be easy to follow, but remember that we are in $\mathbb{Z}_5[x]$, so, for example, $4x - (-3x) = 2x$.

$$\begin{array}{r|l} & x^2 - x - 3 \\ \hline x^2 - 2x + 3 & x^4 - 3x^3 + 2x^2 + 4x - 1 \\ & \underline{x^4 - 2x^3 + 3x^2} \\ & \quad - x^3 - x^2 + 4x \\ & \quad \underline{- x^3 + 2x^2 - 3x} \\ & \qquad - 3x^2 + 2x - 1 \\ & \qquad \underline{- 3x^2 + x - 4} \\ & \qquad\qquad x + 3 \end{array}$$

Thus

$$q(x) = x^2 - x - 3, \quad \text{and} \quad r(x) = x + 3. \qquad \blacktriangle$$

We give three important corollaries of Theorem 23.1. The first one appears in high school algebra for the special case $F[x] = \mathbb{R}[x]$. We phrase our proof in terms of the mapping (homomorphism) approach described in Section 22.

23.3 Corollary **(Factor Theorem)** An element $a \in F$ is a zero of $f(x) \in F[x]$ if and only if $x - a$ is a factor of $f(x)$ in $F[x]$.

Proof Suppose that for $a \in F$ we have $f(a) = 0$. By Theorem 23.1, there exist $q(x)$, $r(x) \in F[x]$ such that

$$f(x) = (x - a)q(x) + r(x),$$

where either $r(x) = 0$ or the degree of $r(x)$ is less than 1. Thus we must have $r(x) = c$ for $c \in F$, so

$$f(x) = (x - a)q(x) + c.$$

Applying our evaluation homomorphism, $\phi_a : F[x] \to F$ of Theorem 22.4, we find

$$0 = f(a) = 0q(a) + c,$$

so it must be that $c = 0$. Then $f(x) = (x - a)q(x)$, so $x - a$ is a factor of $f(x)$.

Conversely, if $x - a$ is a factor of $f(x)$ in $F[x]$, where $a \in F$, then applying our evaluation homomorpohism ϕ_a to $f(x) = (x - a)q(x)$, we have $f(a) = 0q(a) = 0$. ◆

23.4 Example Working again in $\mathbb{Z}_5[x]$, note that 1 is a zero of

$$(x^4 + 3x^3 + 2x + 4) \in \mathbb{Z}_5[x].$$

Thus by Corollary 23.3, we should be able to factor $x^4 + 3x^3 + 2x + 4$ into $(x - 1)q(x)$ in $\mathbb{Z}_5[x]$. Let us find the factorization by long division.

$$\begin{array}{r|l}
 & x^3 + 4x^2 + 4x + 1 \\
\cline{2-2}
x - 1 & x^4 + 3x^3 + \quad\quad 2x + 4 \\
 & \underline{x^4 - x^3} \\
 & \quad 4x^3 \\
 & \quad \underline{4x^3 - 4x^2} \\
 & \quad\quad 4x^2 + 2x \\
 & \quad\quad \underline{4x^2 - 4x} \\
 & \quad\quad\quad x + 4 \\
 & \quad\quad\quad \underline{x - 1} \\
 & \quad\quad\quad\quad 0
\end{array}$$

Thus $x^4 + 3x^3 + 2x + 4 = (x - 1)(x^3 + 4x^2 + 4x + 1)$ in $\mathbb{Z}_5[x]$. Since 1 is seen to be a zero of $x^3 + 4x^2 + 4x + 1$ also, we can divide this polynomial by $x - 1$ and get

$$\begin{array}{r|l}
 & x^2 + 4 \\
\cline{2-2}
x - 1 & x^3 + 4x^2 + 4x + 1 \\
 & \underline{x^3 - x^2} \\
 & \quad 0 \quad + 4x + 1 \\
 & \quad\quad\quad \underline{4x - 4} \\
 & \quad\quad\quad\quad 0
\end{array}$$

Since $x^2 + 4$ still has 1 as a zero, we can divide again by $x - 1$ and get

$$\begin{array}{r|l}
 & x + 1 \\
\cline{2-2}
x - 1 & x^2 \quad\quad + 4 \\
 & \underline{x^2 - x} \\
 & \quad\quad x + 4 \\
 & \quad\quad \underline{x - 1} \\
 & \quad\quad\quad 0
\end{array}$$

Thus $x^4 + 3x^3 + 2x + 4 = (x - 1)^3(x + 1)$ in $\mathbb{Z}_5[x]$. ▲

The next corollary should also look familiar.

23.5 Corollary A nonzero polynomial $f(x) \in F[x]$ of degree n can have at most n zeros in a field F.

Proof The preceding corollary shows that if $a_1 \in F$ is a zero of $f(x)$, then

$$f(x) = (x - a_1)q_1(x),$$

where, of course, the degree of $q_1(x)$ is $n - 1$. A zero $a_2 \in F$ of $q_1(x)$ then results in a factorization

$$f(x) = (x - a_1)(x - a_2)q_2(x).$$

Continuing this process, we arrive at

$$f(x) = (x - a_1) \cdots (x - a_r)q_r(x),$$

where $q_r(x)$ has no further zeros in F. Since the degree of $f(x)$ is n, at most n factors $(x - a_i)$ can appear on the right-hand side of the preceding equation, so $r \leq n$. Also, if $b \neq a_i$ for $i = 1, \cdots, r$ and $b \in F$, then

$$f(b) = (b - a_1) \cdots (b - a_r)q_r(b) \neq 0,$$

since F has no divisors of 0 and none of $b - a_i$ or $q_r(b)$ are 0 by construction. Hence the a_i for $i = 1, \cdots, r \leq n$ are all the zeros in F of $f(x)$. ◆

Our final corollary is concerned with the structure of the multiplicative group F^* of nonzero elements of a field F, rather than with factorization in $F[x]$. It may at first seem surprising that such a result follows from the division algorithm in $F[x]$, but recall that the result that a subgroup of a cyclic group is cyclic follows from the division algorithm in $\mathbb{Z}$.

23.6 Corollary If G is a finite subgroup of the multiplicative group $\langle F^*, \cdot \rangle$ of a field F, then G is cyclic. In particular, the multiplicative group of all nonzero elements of a finite field is cyclic.

Proof By Theorem 11.12 as a finite abelian group, G is isomorphic to a direct product $\mathbb{Z}_{d_1} \times \mathbb{Z}_{d_2} \times \cdots \times \mathbb{Z}_{d_r}$, where each d_i is a power of a prime. Let us think of each of the $\mathbb{Z}_{d_i}$ as a cyclic group of order d_i in *multiplicative* notation. Let m be the least common multiple of all the d_i for $i = 1, 2, \cdots, r$; note that $m \leq d_1 d_2 \cdots d_r$. If $a_i \in \mathbb{Z}_{d_i}$, then $a_i^{d_i} = 1$, so $a_i^m = 1$ since d_i divides m. Thus for all $\alpha \in G$, we have $\alpha^m = 1$, so every element of G is zero of $x^m - 1$. But G has $d_1 d_2 \cdots d_r$ elements, while $x^m - 1$ can have at most m zeros in the field F by Corollary 23.5, so $m \geq d_1 d_2 \cdots d_r$. Hence $m = d_1 d_2 \cdots d_r$, so the primes involved in the prime powers $d_1, d_2, \cdots, d_r$ are distinct, and the group G is isomorphic to the cyclic group $\mathbb{Z}_m$. ◆

Exercises 5 through 8 ask us to find all generators of the cyclic groups of units for some finite fields. The fact that the multiplicative group of units of a finite field is cyclic has been applied in algebraic coding.

Irreducible Polynomials

Our next definition singles out a type of polynomial in $F[x]$ that will be of utmost importance to us. The concept is probably already familiar. We really *are* doing high school algebra in a more general setting.

23.7 Definition A nonconstant polynomial $f(x) \in F[x]$ is **irreducible over** F or is an **irreducible polynomial in** $F[x]$ if $f(x)$ cannot be expressed as a product $g(x)h(x)$ of two polynomials $g(x)$ and $h(x)$ in $F[x]$ both of lower degree than the degree of $f(x)$. If $f(x) \in F[x]$ is a nonconstant polynomial that is not irreducible over F, then $f(x)$ is **reducible over** F. ■

Note that the preceding definition concerns the concept *irreducible over* F and not just the concept *irreducible*. A polynomial $f(x)$ may be irreducible over F, but may not be irreducible if viewed over a larger field E containing F. We illustrate this.

23.8 Example Theorem 22.11 shows that $x^2 - 2$ viewed in $\mathbb{Q}[x]$ has no zeros in $\mathbb{Q}$. This shows that $x^2 - 2$ is irreducible over $\mathbb{Q}$, for a factorization $x^2 - 2 = (ax + b)(cx + d)$ for $a, b, c, d \in \mathbb{Q}$ would give rise to zeros of $x^2 - 2$ in $\mathbb{Q}$. However, $x^2 - 2$ viewed in $\mathbb{R}[x]$ is not irreducible over $\mathbb{R}$, because $x^2 - 2$ factors in $\mathbb{R}[x]$ into $(x - \sqrt{2})(x + \sqrt{2})$. ▲

It is worthwhile to remember that *the units in* $F[x]$ *are precisely the nonzero elements of* F. Thus we could have defined an irreducible polynomial $f(x)$ as a nonconstant polynomial such that in any factorization $f(x) = g(x)h(x)$ in $F[x]$, either $g(x)$ or $h(x)$ is a unit.

23.9 Example Let us show that $f(x) = x^3 + 3x + 2$ viewed in $\mathbb{Z}_5[x]$ is irreducible over $\mathbb{Z}_5$. If $x^3 + 3x + 2$ factored in $\mathbb{Z}_5[x]$ into polynomials of lower degree then there would exist at least one linear factor of $f(x)$ of the form $x - a$ for some $a \in \mathbb{Z}_5$. But then $f(a)$ would be 0, by Corollary 23.3. However, $f(0) = 2$, $f(1) = 1$, $f(-1) = -2$, $f(2) = 1$, and $f(-2) = -2$, showing that $f(x)$ has no zeros in $\mathbb{Z}_5$. Thus $f(x)$ is irreducible over $\mathbb{Z}_5$. This test for irreducibility by finding zeros works nicely for quadratic and cubic polynomials over a finite field with a small number of elements. ▲

Irreducible polynomials will play a very important role in our work from now on. The problem of determining whether a given $f(x) \in F[x]$ is irreducible over F may be difficult. We now give some criteria for irreducibility that are useful in certain cases. One technique for determining irreducibility of quadratic and cubic polynomials was illustrated in Examples 23.8 and 23.9. We formalize it in a theorem.

23.10 Theorem Let $f(x) \in F[x]$, and let $f(x)$ be of degree 2 or 3. Then $f(x)$ is reducible over F if and only if it has a zero in F.

Proof If $f(x)$ is reducible so that $f(x) = g(x)h(x)$, where the degree of $g(x)$ and the degree of $h(x)$ are both less than the degree of $f(x)$, then since $f(x)$ is either quadratic or cubic, either $g(x)$ or $h(x)$ is of degree 1. If, say, $g(x)$ is of degree 1, then except for a possible factor in F, $g(x)$ is of the form $x - a$. Then $g(a) = 0$, which implies that $f(a) = 0$, so $f(x)$ has a zero in F.

Conversely, Corollary 23.3 shows that if $f(a) = 0$ for $a \in F$, then $x - a$ is a factor of $f(x)$, so $f(x)$ is reducible. ◆

We turn to some conditions for irreducibility over $\mathbb{Q}$ of polynomials in $\mathbb{Q}[x]$. The most important condition that we shall give is contained in the next theorem. We shall not prove this theorem here; it involves clearing denominators and gets a bit messy.

23.11 Theorem If $f(x) \in \mathbb{Z}[x]$, then $f(x)$ factors into a product of two polynomials of lower degrees r and s in $\mathbb{Q}[x]$ if and only if it has such a factorization with polynomials of the same degrees r and s in $\mathbb{Z}[x]$.

Proof The proof is omitted here. ◆

23.12 Corollary If $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$ is in $\mathbb{Z}[x]$ with $a_0 \neq 0$, and if $f(x)$ has a zero in $\mathbb{Q}$, then it has a zero m in $\mathbb{Z}$, and m must divide a_0.

Proof If $f(x)$ has a zero a in $\mathbb{Q}$, then $f(x)$ has a linear factor $x - a$ in $\mathbb{Q}[x]$ by Corollary 23.3. But then by Theorem 23.11, $f(x)$ has a factorization with a linear factor in $\mathbb{Z}[x]$, so for some $m \in \mathbb{Z}$ we must have

$$f(x) = (x - m)\big(x^{n-1} + \cdots - a_0/m\big).$$

Thus a_0/m is in $\mathbb{Z}$, so m divides a_0. ◆

23.13 Example Corollary 23.12 gives us another proof of the irreducibility of $x^2 - 2$ over $\mathbb{Q}$, for $x^2 - 2$ factors nontrivially in $\mathbb{Q}[x]$ if and only if it has a zero in $\mathbb{Q}$ by Theorem 23.10. By Corollary 23.12, it has a zero in $\mathbb{Q}$ if and only if it has a zero in $\mathbb{Z}$, and moreover the only possibilities are the divisors ± 1 and ± 2 of 2. A check shows that none of these numbers is a zero of $x^2 - 2$. ▲

23.14 Example Let us use Theorem 23.11 to show that

$$f(x) = x^4 - 2x^2 + 8x + 1$$

viewed in $\mathbb{Q}[x]$ is irreducible over $\mathbb{Q}$. If $f(x)$ has a linear factor in $\mathbb{Q}[x]$, then it has a zero in $\mathbb{Z}$, and by Corollary 23.12, this zero would have to be a divisor in $\mathbb{Z}$ of 1, that is, either ± 1. But $f(1) = 8$, and $f(-1) = -8$, so such a factorization is impossible.

If $f(x)$ factors into two quadratic factors in $\mathbb{Q}[x]$, then by Theorem 23.11, it has a factorization.

$$(x^2 + ax + b)(x^2 + cx + d)$$

in $\mathbb{Z}[x]$. Equating coefficients of powers of x, we find that we must have

$$bd = 1, \quad ad + bc = 8, \quad ac + b + d = -2, \quad \text{and} \quad a + c = 0$$

for integers $a, b, c, d \in \mathbb{Z}$. From $bd = 1$, we see that either $b = d = 1$ or $b = d = -1$. In any case, $b = d$ and from $ad + bc = 8$, we deduce that $d(a + c) = 8$. But this is impossible since $a + c = 0$. Thus a factorization into two quadratic polynomials is also impossible and $f(x)$ is irreducible over $\mathbb{Q}$. ▲

We conclude our irreducibility criteria with the famous Eisenstein criterion for irreducibility. An additional very useful criterion is given in Exercise 37.

23.15 Theorem **(Eisenstein Criterion)** Let $p \in \mathbb{Z}$ be a prime. Suppose that $f(x) = a_n x^n + \cdots + a_0$ is in $\mathbb{Z}[x]$, and $a_n \not\equiv 0 \pmod{p}$, but $a_i = 0 \pmod{p}$ for all $i < n$, with $a_0 \not\equiv 0 \pmod{p^2}$. Then $f(x)$ is irreducible over $\mathbb{Q}$.

Proof By Theorem 23.11 we need only show that $f(x)$ does not factor into polynomials of lower degree in $\mathbb{Z}[x]$. If

$$f(x) = (b_r x^r + \cdots + b_0)(c_s x^s + \cdots + c_0)$$

is a factorization in $\mathbb{Z}[x]$, with $b_r \not\equiv 0$, $c_s \not\equiv 0$ and $r, s < n$, then $a_0 \not\equiv 0 \pmod{p^2}$ implies that b_0 and c_0 are not both congruent to 0 modulo p. Suppose that $b_0 \not\equiv 0 \pmod{p}$ and $c_0 \equiv 0 \pmod{p}$. Now $a_n \not\equiv 0 \pmod{p}$ implies that $b_r, c_s \not\equiv 0 \pmod{p}$, since $a_n = b_r c_s$. Let m be the smallest value of k such that $c_k \not\equiv 0 \pmod{p}$. Then

$$a_m = b_0 c_m + b_1 c_{m-1} + \cdots + \begin{cases} b_m c_0 \text{ if } r \geq m, \\ b_r c_{m-r} \text{ if } r < m. \end{cases}$$

The fact that neither b_0 nor c_m are congruent to 0 modulo p while $c_{m-1}, \cdots, c_0$ are all congruent to 0 modulo p implies that $a_m \not\equiv 0$ modulo p, so $m = n$. Consequently, $s = n$, contradicting our assumption that $s < n$; that is, that our factorization was nontrivial. ◆

Note that if we take $p = 2$, the Eisenstein criterion gives us still another proof of the irreducibility of $x^2 - 2$ over $\mathbb{Q}$.

23.16 Example Taking $p = 3$, we see by Theorem 23.15 that

$$25x^5 - 9x^4 - 3x^2 - 12$$

is irreducible over $\mathbb{Q}$. ▲

23.17 Corollary The polynomial

$$\Phi_p(x) = \frac{x^p - 1}{x - 1} = x^{p-1} + x^{p-2} + \cdots + x + 1$$

is irreducible over $\mathbb{Q}$ for any prime p.

Proof Again by Theorem 23.11, we need only consider factorizations in $\mathbb{Z}[x]$. We remarked following Theorem 22.5 that its proof actually shows that evaluation homomorphims can be used for commutative rings. Here we want to use the evaluation homomorphism $\phi_{x+1} : \mathbb{Q}[x] \to \mathbb{Q}[x]$. It is natural for us to denote $\phi_{x+1}(f(x))$ by $f(x+1)$ for $f(x) \in \mathbb{Q}[x]$. Let

$$g(x) = \Phi_p(x+1) = \frac{(x+1)^p - 1}{(x+1) - 1} = \frac{x^p + \binom{p}{1} x^{p-1} + \cdots + px}{x}.$$

The coefficient of x^{p-r} for $0 < r < p$ is the binomial coefficient $p!/[r!(p-r)!]$ which is divisible by p because p divides $p!$ but does not divide either $r!$ or $(p-r)!$ when $0 < r < p$. Thus

$$g(x) = x^{p-1} + \binom{p}{1} x^{p-2} + \cdots + p$$

satisifies the Eisenstein criterion for the prime p and is thus irreducible over $\mathbb{Q}$. But if $\Phi_p(x) = h(x)r(x)$ were a nontrivial factorization of $\Phi_p(x)$ in $\mathbb{Z}[x]$, then

$$\Phi_p(x+1) = g(x) = h(x+1)r(x+1)$$

would give a nontrivial factorization of $g(x)$ in $\mathbb{Z}[x]$. Thus $\Phi_p(x)$ must also be irreducible over $\mathbb{Q}$. ◆

The polynomial $\Phi_p(x)$ in Corollary 23.17 is the **p^{th} cyclotomic polynomial.**

Uniqueness of Factorization in $F[x]$

Polynomials in $F[x]$ can be factored into a product of irreducible polynomials in $F[x]$ in an essentially unique way. For $f(x), g(x) \in F[x]$ we say that $g(x)$ **divides** $f(x)$ **in** $F[x]$ if there exists $q(x) \in F[x]$ such that $f(x) = g(x)q(x)$. Note the similarity of the theorem that follows with boxed Property (1) for $\mathbb{Z}$ following Example 6.9.

23.18 Theorem Let $p(x)$ be an irreducible polynomial in $F[x]$. If $p(x)$ divides $r(x)s(x)$ for $r(x), s(x) \in F[x]$, then either $p(x)$ divides $r(x)$ or $p(x)$ divides $s(x)$.

Proof We delay the proof of this theorem to Section 27. (See Theorem 27.27.) ◆

23.19 Corollary If $p(x)$ is irreducible in $F[x]$ and $p(x)$ divides the product $r_1(x) \cdots r_n(x)$ for $r_i(x) \in F[x]$, then $p(x)$ divides $r_i(x)$ for at least one i.

Proof Using mathematical induction, we find that this is immediate from Theorem 23.18. ◆

23.20 Theorem If F is a field, then every nonconstant polynomial $f(x) \in F[x]$ can be factored in $F[x]$ into a product of irreducible polynomials, the irreducible polynomials being unique except for order and for unit (that is, nonzero constant) factors in F.

Proof Let $f(x) \in F[x]$ be a nonconstant polynomial. If $f(x)$ is not irreducible, then $f(x) = g(x)h(x)$, with the degree of $g(x)$ and the degree of $h(x)$ both less than the degree of $f(x)$. If $g(x)$ and $h(x)$ are both irreducible, we stop here. If not, at least one of them factors into polynomials of lower degree. Continuing this process, we arrive at a factorization

$$f(x) = p_1(x)p_2(x) \cdots p_r(x),$$

where $p_i(x)$ is irreducible for $i = 1, 2, \cdots, r$.

It remains for us to show uniqueness. Suppose that

$$f(x) = p_1(x)p_2(x) \cdots p_r(x) = q_1(x)q_2(x) \cdots q_s(x)$$

are two factorizations of $f(x)$ into irreducible polynomials. Then by Corollary 23.19, $p_1(x)$ divides some $q_j(x)$, let us assume $q_1(x)$. Since $q_1(x)$ is irreducible,

$$q_1(x) = u_1 p_1(x),$$

where $u_1 \neq 0$, but u_1 is in F and thus is a unit. Then substituting $u_1 p_1(x)$ for $q_1(x)$ and canceling, we get

$$p_2(x) \cdots p_r(x) = u_1 q_2(x) \cdots q_s(x).$$

By a similar argument, say $q_2(x) = u_2 p_2(x)$, so

$$p_3(x) \cdots p_r(x) = u_1 u_2 q_3(x) \cdots q_s(x).$$

Continuing in this manner, we eventually arrive at

$$1 = u_1 u_2 \cdots u_r q_{r+1}(x) \cdots q_s(x).$$

This is only possible if $s = r$, so that this equation is actually $1 = u_1 u_2 \cdots u_r$. Thus the irreducible factors $p_i(x)$ and $q_j(x)$ were the same except possibly for order and unit factors. ◆

23.21 Example Example 23.4 shows a factorization of $x^4 + 3x^3 + 2x + 4$ in $\mathbb{Z}_5[x]$ is $(x-1)^3(x+1)$. These irreducible factors in $\mathbb{Z}_5[x]$ are only unique up to units in $\mathbb{Z}_5[x]$, that is, nonzero constants in $\mathbb{Z}_5$. For example, $(x-1)^3(x+1) = (x-1)^2(2x-2)(3x+3)$. ▲

EXERCISES 23

Computations

In Exercises 1 through 4, find $q(x)$ and $r(x)$ as described by the division algorithm so that $f(x) = g(x)q(x) + r(x)$ with $r(x) = 0$ or of degree less than the degree of $g(x)$.

1. $f(x) = x^6 + 3x^5 + 4x^2 - 3x + 2$ and $g(x) = x^2 + 2x - 3$ in $\mathbb{Z}_7[x]$.
2. $f(x) = x^6 + 3x^5 + 4x^2 - 3x + 2$ and $g(x) = 3x^2 + 2x - 3$ in $\mathbb{Z}_7[x]$.
3. $f(x) = x^5 - 2x^4 + 3x - 5$ and $g(x) = 2x + 1$ in $\mathbb{Z}_{11}[x]$.
4. $f(x) = x^4 + 5x^3 - 3x^2$ and $g(x) = 5x^2 - x + 2$ in $\mathbb{Z}_{11}[x]$.

In Exercises 5 through 8, find all generators of the cyclic multiplicative group of units of the given finite field. (Review Corollary 6.16.)

5. $\mathbb{Z}_5$ 6. $\mathbb{Z}_7$ 7. $\mathbb{Z}_{17}$ 8. $\mathbb{Z}_{23}$

9. The polynomial $x^4 + 4$ can be factored into linear factors in $\mathbb{Z}_5[x]$. Find this factorization.
10. The polynomial $x^3 + 2x^2 + 2x + 1$ can be factored into linear factors in $\mathbb{Z}_7[x]$. Find this factorization.
11. The polynomial $2x^3 + 3x^2 - 7x - 5$ can be factored into linear factors in $\mathbb{Z}_{11}[x]$. Find this factorization.
12. Is $x^3 + 2x + 3$ an irreducible polynomial of $\mathbb{Z}_5[x]$? Why? Express it as a product of irreducible polynomials of $\mathbb{Z}_5[x]$.
13. Is $2x^3 + x^2 + 2x + 2$ an irreducible polynomial in $\mathbb{Z}_5[x]$? Why? Express it as a product of irreducible polynomials in $\mathbb{Z}_5[x]$.
14. Show that $f(x) = x^2 + 8x - 2$ is irreducible over $\mathbb{Q}$. Is $f(x)$ irreducible over $\mathbb{R}$? Over $\mathbb{C}$?
15. Repeat Exercise 14 with $g(x) = x^2 + 6x + 12$ in place of $f(x)$.
16. Demonstrate that $x^3 + 3x^2 - 8$ is irreducible over $\mathbb{Q}$.
17. Demonstrate that $x^4 - 22x^2 + 1$ is irreducible over $\mathbb{Q}$.

In Exercises 18 through 21, determine whether the polynomial in $\mathbb{Z}[x]$ satisfies an Eisenstein criterion for irreducibility over $\mathbb{Q}$.

18. $x^2 - 12$

19. $8x^3 + 6x^2 - 9x + 24$

20. $4x^{10} - 9x^3 + 24x - 18$

21. $2x^{10} - 25x^3 + 10x^2 - 30$

22. Find all zeros of $6x^4 + 17x^3 + 7x^2 + x - 10$ in $\mathbb{Q}$. (This is a tedious high school algebra problem. *You* might use a bit of analytic geometry and calculus and make a graph, or use Newton's method to see which are the best candidates for zeros.)

Concepts

In Exercises 23 and 24, correct the definition of the italicized term without reference to the text, if correction is needed, so that it is in a form acceptable for publication.

23. A polynomial $f(x) \in F[x]$ is *irreducible over the field* F if and only if $f(x) \neq g(x)h(x)$ for any polynomials $g(x), h(x) \in F[x]$.

24. A nonconstant polynomial $f(x) \in F[x]$ is *irreducible over the field* F if and only if in any factorization of it in $F[x]$, one of the factors is in F.

25. Mark each of the following true or false.

_____ **a.** $x - 2$ is irreducible over $\mathbb{Q}$.

_____ **b.** $3x - 6$ is irreducible over $\mathbb{Q}$.

_____ **c.** $x^2 - 3$ is irreducible over $\mathbb{Q}$.

_____ **d.** $x^2 + 3$ is irreducible over $\mathbb{Z}_7$.

_____ **e.** If F is a field, the units of $F[x]$ are precisely the nonzero elements of F.

_____ **f.** If F is a field, the units of $F[x]$ are precisely the nonzero elements of F.

_____ **g.** A polynomial $f(x)$ of degree n with coefficients in a field F can have at most n zeros in F.

_____ **h.** A polynomial $f(x)$ of degree n with coefficients in a field F can have at most n zeros in any given field E such that $F \leq E$.

_____ **i.** Every polynomial of degree 1 in $F[x]$ has at least one zero in the field F.

_____ **j.** Each polynomial in $F[x]$ can have at most a finite number of zeros in the field F.

26. Find all prime numbers p such that $x + 2$ is a factor of $x^4 + x^3 + x^2 - x + 1$ in $\mathbb{Z}_p[x]$.

In Exercises 27 through 30, find all irreducible polynomials of the indicated degree in the given ring.

27. Degree 2 in $\mathbb{Z}_2[x]$

28. Degree 3 in $\mathbb{Z}_2[x]$

29. Degree 2 in $\mathbb{Z}_3[x]$

30. Degree 3 in $\mathbb{Z}_3[x]$

31. Find the number of irreducible quadratic polynomials in $\mathbb{Z}_p[x]$, where p is a prime. [*Hint:* Find the number of reducible polynomials of the form $x^2 + ax + b$, then the number of reducible quadratics, and subtract this from the total number of quadratics.]

Proof Synopsis

32. Give a synopsis of the proof of Corollary 23.5.

33. Give a synopsis of the proof of Corollary 23.6.

Theory

34. Show that for p a prime, the polynomial $x^p + a$ in $\mathbb{Z}_p[x]$ is not irreducible for any $a \in \mathbb{Z}_p$.

35. If F is a field and $a \neq 0$ is a zero of $f(x) = a_0 + a_1x + \cdots + a_nx^n$ in $F[x]$, show that $1/a$ is a zero of $a_n + a_{n-1}x + \cdots + a_0x^n$.

36. (Remainder Theorem) Let $f(x) \in F[x]$ where F is a field, and let $\alpha \in F$. Show that the remainder $r(x)$ when $f(x)$ is divided by $x - \alpha$, in accordance with the division algorithm, is $f(\alpha)$.

37. Let $\sigma_m : \mathbb{Z} \to \mathbb{Z}_m$ be the natural homomorphism given by $\sigma_m(a) =$ (the remainder of a when divided by m) for $a \in \mathbb{Z}$.

a. Show that $\overline{\sigma_m} : \mathbb{Z}[x] \to \mathbb{Z}_m[x]$ given by

$$\overline{\sigma_m}(a_0 + a_1x + \cdots + a_nx^n) = \sigma_m(a_0) + \sigma_m(a_1)x + \cdots + \sigma_m(a_n)x^n$$

is a homomorphism of $\mathbb{Z}[x]$ onto $\mathbb{Z}_m[x]$.

b. Show that if $f(x) \in \mathbb{Z}[x]$ and $\overline{\sigma_m}(f(x))$ both have degree n and $\overline{\sigma_m}(f(x))$ does not factor in $\mathbb{Z}_m[x]$ into two polynomials of degree less than n, then $f(x)$ is irreducible in $\mathbb{Q}[x]$.

c. Use part (b) to show that $x^3 + 17x + 36$ is irreducible in $\mathbb{Q}[x]$. [*Hint:* Try a prime value of m that simplifies the coefficients.]

SECTION 24 †NONCOMMUTATIVE EXAMPLES

Thus far, the only example we have presented of a ring that is not commutative is the ring $M_n(F)$ of all $n \times n$ matrices with entries in a field F. We shall be doing almost nothing with noncommutative rings and strictly skew fields. To show that there are other important noncommutative rings occurring very naturally in algebra, we give several examples of such rings.

Rings of Endomorphisms

Let A be any abelian group. A homomorphism of A into itself is an **endomorphism of** A. Let the set of all endomorphisms of A be $\text{End}(A)$. Since the composition of two homomorphisms of A into itself is again such a homomorphism, we define multiplication on $\text{End}(A)$ by function composition, and thus multiplication is associative.

To define addition, for $\phi, \psi \in \text{End}(A)$, we have to describe the value of $(\phi + \psi)$ on each $a \in A$. Define

$$(\phi + \psi)(a) = \phi(a) + \psi(a).$$

Since

$$\begin{aligned}(\phi + \psi)(a + b) &= \phi(a + b) + \psi(a + b)\\ &= [\phi(a) + \phi(b)] + [\psi(a) + \psi(b)]\\ &= [\phi(a) + \psi(a)] + [\phi(b) + \psi(b)]\\ &= (\phi + \psi)(a) + (\phi + \psi)(b)\end{aligned}$$

we see that $\phi + \psi$ is again in $\text{End}(A)$.

Since A is commutative, we have

$$(\phi + \psi)(a) = \phi(a) + \psi(a) = \psi(a) + \phi(a) = (\psi + \phi)(a)$$

for all $a \in A$, so $\phi + \psi = \psi + \phi$ and addition in $\text{End}(A)$ is commutative. The associativity of addition follows from

† This section is not used in the remainder of the text.

$$\begin{aligned}[\phi + (\psi + \theta)](a) &= \phi(a) + [(\psi + \theta)(a)] \\ &= \phi(a) + [\psi(a) + \theta(a)] \\ &= [\phi(a) + \psi(a)] + \theta(a) \\ &= (\phi + \psi)(a) + \theta(a) \\ &= [(\phi + \psi) + \theta](a).\end{aligned}$$

If e is the additive identity of A, then the homomorphism 0 defined by

$$0(a) = e$$

for $a \in A$ is an additive identity in $\text{End}(A)$. Finally, for

$$\phi \in \text{End}(A),$$

$-\phi$ defined by

$$(-\phi)(a) = -\phi(a)$$

is in $\text{End}(A)$, since

$$\begin{aligned}(-\phi)(a + b) &= -\phi(a + b) = -[\phi(a) + \phi(b)] \\ &= -\phi(a) - \phi(b) = (-\phi)(a) + (-\phi)(b),\end{aligned}$$

and $\phi + (-\phi) = 0$. Thus $\langle \text{End}(A), + \rangle$ is an abelian group.

Note that we have not yet used the fact that our functions are *homomorphisms* except to show that $\phi + \psi$ and $-\phi$ are again *homomorphisms*. Thus the set A^A of *all functions* from A into A is an abelian group under exactly the same definition of addition, and, of course, function composition again gives a nice associative multiplication in A^A. However, we do need the fact that these functions in $\text{End}(A)$ are homomorphisms now to prove the left distributive law in $\text{End}(A)$. Except for this left distributive law, $\langle A^A, +, \cdot \rangle$ satisfies all the axioms for a ring. Let ϕ, ψ, and θ be in $\text{End}(A)$, and let $a \in A$. Then

$$(\theta(\phi + \psi))(a) = \theta((\phi + \psi)(a)) = \theta(\phi(a) + \psi(a)).$$

Since θ is a *homomorphism,*

$$\begin{aligned}\theta(\phi(a) + \psi(a)) &= \theta(\phi(a)) + \theta(\psi(a)) \\ &= (\theta\phi)(a) + (\theta\psi)(a) \\ &= (\theta\phi + \theta\psi)(a).\end{aligned}$$

Thus $\theta(\phi + \psi) = \theta\phi + \theta\psi$. The right distributive law causes no trouble, even in A^A, and follows from

$$\begin{aligned}((\psi + \theta)\phi)(a) &= (\psi + \theta)(\phi(a)) = \psi(\phi(a)) + \theta(\phi(a)) \\ &= (\psi\phi)(a) + (\theta\phi)(a) = (\psi\phi + \theta\phi)(a).\end{aligned}$$

Thus we have proved the following theorem.

24.1 Theorem The set $\text{End}(A)$ of all endomorphisms of an abelian group A forms a ring under homomorphism addition and homomorphism multiplication (function composition).

Again, to show relevance to this section, we should give an example showing that $\text{End}(A)$ need not be commutative. Since function composition is in general not commutative, this seems reasonable to expect. However, $\text{End}(A)$ may be commutative in some cases. Indeed, Exercise 15 asks us to show that $\text{End}(\langle \mathbb{Z}, + \rangle)$ is commutative.

24.2 Example Consider the abelian group $\langle \mathbb{Z} \times \mathbb{Z}, + \rangle$ discussed in Section 11. It is straightforward to verify that two elements of $\text{End}(\langle \mathbb{Z} \times \mathbb{Z}, + \rangle)$ are ϕ and ψ defined by

$$\phi((m, n)) = (m + n, 0) \qquad \text{and} \qquad \psi((m, n)) = (0, n).$$

Note that ϕ maps everything onto the first factor of $\mathbb{Z} \times \mathbb{Z}$, and ψ collapses the first factor. Thus

$$(\psi\phi)(m, n) = \psi(m + n, 0) = (0, 0).$$

while

$$(\phi\psi)(m, n) = \phi(0, n) = (n, 0).$$

Hence $\phi\psi \neq \psi\phi$. ▲

24.3 Example Let F be a field of characteristic zero, and let $\langle F[x], + \rangle$ be the additive group of the ring $F[x]$ of polynomials with coefficients in F. For this example, let us denote this additive group by $F[x]$, to simplify this notation. We can consider $\text{End}(F[x])$. One element of $\text{End}(F[x])$ acts on each polynomial in $F[x]$ by multiplying it by x. Let this endomorphism be X, so

$$X(a_0 + a_1x + a_2x^2 + \cdots + a_nx^n) = a_0x + a_1x^2 + a_2x^3 + \cdots + a_nx^{n+1}.$$

Another element of $\text{End}(F[x])$ is formal differentiation with respect to x. (The familiar formula "the derivation of a sum is the sum of the derivatives" guarantees that differentiation is an endomorphism of $F[x]$.) Let Y be this endomorphism, so

$$Y(a_0 + a_1x + a_2x^2 + \cdots + a_nx^n) = a_1 + 2a_2x + \cdots + na_nx^{n-1}.$$

Exercise 17 asks us to show that $YX - XY = 1$, where 1 is unity (the identity map) in $\text{End}(F[x])$. Thus $XY \neq YX$. Multiplication of polynomials in $F[x]$ by any element of F also gives an element of $\text{End}(F[x])$. The subring of $\text{End}(F[x])$ generated by X and Y and multiplications by elements of F is the **Weyl algebra** and is important in quantum mechanics. ▲

Group Rings and Group Algebras

Let $G = \{g_i \mid i \in I\}$ be any group written multiplicatively and let R be any commutative ring with nonzero unity. Let RG be the set of all *formal sums*.

$$\sum_{i \in I} a_i g_i$$

for $a_i \in R$ and $g_i \in G$, *where all but a finite number of the* a_i *are 0*. Define the sum of two elements of RG by

$$\left(\sum_{i \in I} a_i g_i\right) + \left(\sum_{i \in I} b_i g_i\right) = \sum_{i \in I} (a_i + b_i) g_i.$$

Observe that $(a_i + b_i) = 0$ except for a finite number of indices i, so $\Sigma_{i \in I}(a_i + b_i)g_i$ is again in RG. It is immediate that $\langle RG, + \rangle$ is an abelian group with additive identity $\Sigma_{i \in I} 0 g_i$.

Multiplication of two elements of RG is defined by the use of the multiplications in G and R as follows:

$$\left(\sum_{i\in I} a_i g_i\right)\left(\sum_{i\in I} b_i g_i\right) = \sum_{i\in I}\left(\sum_{g_j g_k = g_i} a_j b_k\right) g_i.$$

Naively, we formally distribute the sum $\Sigma_{i\in I} a_i g_i$ over the sum $\Sigma_{i\in I} b_i g_i$ and rename a term $a_j g_j b_k g_k$ by $a_j b_k g_i$ where $g_j g_k = g_i$ in G. Since a_i and b_i are 0 for all but a finite number of i, the sum $\Sigma_{g_j g_k = g_i} a_j b_k$ contains only a finite number of nonzero summands $a_j b_k \in R$ and may thus be viewed as an element of R. Again, at most a finite number of such sums $\Sigma_{g_j g_k = g_i} a_j b_k$ are nonzero. Thus multiplication is closed on RG.

The distributive laws follow at once from the definition of addition and the formal way we used distributivity to define multiplication. For the associativity of multiplication

$$\begin{aligned}\left(\sum_{i\in I} a_i g_i\right)\left[\left(\sum_{i\in I} b_i g_i\right)\left(\sum_{i\in I} c_i g_i\right)\right] &= \left(\sum_{i\in I} a_i g_i\right)\left[\sum_{i\in I}\left(\sum_{g_j g_k = g_i} b_j c_k\right) g_i\right] \\ &= \sum_{i\in I}\left(\sum_{g_h g_j g_k = g_i} a_h b_j c_k\right) g_i \\ &= \left[\sum_{i\in I}\left(\sum_{g_h g_j = g_i} a_h b_j\right) g_i\right]\left(\sum_{i\in I} c_i g_i\right) \\ &= \left[\left(\sum_{i\in I} a_i g_i\right)\left(\sum_{i\in I} b_i g_i\right)\right]\left(\sum_{i\in I} c_i g_i\right).\end{aligned}$$

Thus we have proved the following theorem.

24.4 Theorem If G is any group written multiplicatively and R is a commutative ring with nonzero unity, then $\langle RG, +, \cdot\rangle$ is a ring.

Corresponding to each $g \in G$, we have an element $1g$ in RG. If we identify (rename) $1g$ with g, we see that $\langle RG, \cdot\rangle$ can be considered to contain G naturally as a multiplicative subsystem. Thus, if G is not abelian, RG is not a commutative ring.

24.5 Definition The ring RG defined above is the **group ring of G over R**. If F is a field, then FG is the **group algebra of G over F**. ■

24.6 Example Let us give the addition and multiplication tables for the group algebra $\mathbb{Z}_2 G$, where $G = \{e, a\}$ is cyclic of order 2. The elements of $Z_2 G$ are

$$0e + 0a, \quad 0e + 1a, \quad 1e + 0a, \quad \text{and} \quad 1e + 1a.$$

If we denote these elements in the obvious, natural way by

$$0, \quad a, \quad e, \quad \text{and} \quad e + a,$$

24.7 Table

$+$	0	a	e	$e+a$
0	0	a	e	$e+a$
a	a	0	$e+a$	e
e	e	$e+a$	0	a
$e+a$	$e+a$	e	a	0

24.8 Table

	0	a	e	$e+a$
0	0	0	0	0
a	0	e	a	$e+a$
e	0	a	e	$e+a$
$e+a$	0	$e+a$	$e+a$	0

respectively, we get Tables 24.7 and 24.8. For example, to see that $(e + a)(e + a) = 0$, we have

$$(1e + 1a)(1e + 1a) = (1 + 1)e + (1 + 1)a = 0e + 0a.$$

This example shows that a group algebra may have 0 divisors. Indeed, this is usually the case. ▲

The Quaternions

We have not yet given an example of a noncommutative division ring. The *quaternions* of Hamilton are the standard example of a strictly skew field; let us describe them.

▩ Historical Note

Sir William Rowan Hamilton (1805–1865) discovered quaternions in 1843 while he was searching for a way to multiply number triplets (vectors in $\mathbb{R}^3$). Six years earlier he had developed the complex numbers abstractly as pairs (a, b) of real numbers with addition $(a, b) + (a' + b') = (a + a', b + b')$ and multiplication $(a, b)(a'b') = (aa' - bb', ab' + a'b)$; he was then looking for an analogous multiplication for 3-vectors that was distributive and such that the length of the product vector was the product of the lengths of the factors. After many unsuccessful attempts to multiply vectors of the form $a + bi + cj$ (where $1, i, j$ are mutually perpendicular), he realized while walking along the Royal Canal in Dublin on October 16, 1843, that he needed a new "imaginary symbol" k to be perpendicular to the other three elements. He could not "resist the impulse . . . to cut with a knife on a stone of Brougham Bridge" the fundamental defining formulas on page 225 for multiplying these quaternions.

The quaternions were the first known example of a strictly skew field. Though many others were subsequently discovered, it was eventually noted that none were finite. In 1909 Joseph Henry Maclagan Wedderburn (1882–1948), then a preceptor at Princeton University, gave the first proof of Theorem 24.10.

Let the set $\mathbb{H}$, for Hamilton, be $\mathbb{R} \times \mathbb{R} \times \mathbb{R} \times \mathbb{R}$. Now $\langle \mathbb{R} \times \mathbb{R} \times \mathbb{R} \times \mathbb{R}, + \rangle$ is a group under addition by components, the direct product of $\mathbb{R}$ under addition with itself four times. This gives the operation of addition on $\mathbb{H}$. Let us rename certain elements of $\mathbb{H}$. We shall let

$$1 = (1, 0, 0, 0), \quad i = (0, 1, 0, 0),$$
$$j = (0, 0, 1, 0), \quad \text{and} \quad k = (0, 0, 0, 1).$$

We furthermore agree to let

$$a_1 = (a_1, 0, 0, 0), \quad a_2 i = (0, a_2, 0, 0),$$
$$a_3 j = (0, 0, a_3, 0) \quad \text{and} \quad a_4 k = (0, 0, 0, a_4).$$

In view of our definition of addition, we then have

$$(a_1, a_2, a_3, a_4) = a_1 + a_2 i + a_3 j + a_4 k.$$

Thus

$$\begin{aligned}(a_1 + a_2 i + a_3 j + a_4 k) + (b_1 + b_2 i + b_3 j + b_4 k)\\ = (a_1 + b_1) + (a_2 + b_2)i + (a_3 + b_3)j + (a_4 + b_4)k.\end{aligned}$$

To define multiplication on $\mathbb{H}$, we start by defining

$$1a = a1 = a \quad \text{for} \quad a \in \mathbb{H},$$
$$i^2 = j^2 = k^2 = -1,$$

and

$$ij = k, \quad jk = i, \quad ki = j, \quad ji = -k, \quad kj = -i, \quad \text{and} \quad ik = -j.$$

Note the similarity with the so-called cross product of vectors. These formulas are easy to remember if we think of the sequence

$$i, j, k, i, j, k.$$

The product from left to right of two adjacent elements is the next one to the right. The product from right to left of two adjacent elements is the negative of the next one to the left. We then define a product to be what it must be to make the distributive laws hold, namely,

$$\begin{aligned}(a_1 + a_2 i + a_3 j + a_4 k)(b_1 + b_2 i + b_3 j + b_4 k)\\ = (a_1 b_1 - a_2 b_2 - a_3 b_3 - a_4 b_4) + (a_1 b_2 + a_2 b_1 + a_3 b_4 - a_4 b_3)i\\ + (a_1 b_3 - a_2 b_4 + a_3 b_1 + a_4 b_2)j\\ + (a_1 b_4 + a_2 b_3 - a_3 b_2 + a_4 b_1)k.\end{aligned}$$

Exercise 19 shows that the quaternions are isomorphic to a subring of $M_2(\mathbb{C})$, so multiplication is associative. Since $ij = k$ and $ji = -k$, we see that multiplication is not commutative, so $\mathbb{H}$ is definitely not a field. Turning to the existence of multiplicative inverses, let $a = a_1 + a_2 i + a_3 j + a_4 k$, with not all $a_i = 0$. Computation shows that

$$(a_1 + a_2 i + a_3 j + a_4 k)(a_1 - a_2 i - a_3 j - a_4 k) = a_1^2 + a_2^2 + a_3^2 + a_4^2.$$

If we let

$$|a|^2 = a_1^2 + a_2^2 + a_3^2 + a_4^2 \qquad \text{and} \qquad \bar{a} = a_1 - a_2 i - a_3 j - a_4 k,$$

we see that

$$\frac{\bar{a}}{|a|^2} = \frac{a_1}{|a|^2} - \left(\frac{a_2}{|a|^2}\right)i - \left(\frac{a_3}{|a|^2}\right)j - \left(\frac{a_4}{|a|^2}\right)k$$

is a multiplicative inverse for a. We consider that we have demonstrated the following theorem.

24.9 Theorem The quaternions $\mathbb{H}$ form a strictly skew field under addition and multiplication.

Note that $G = \{\pm 1, \pm i, \pm j, \pm k\}$ is a group of order 8 under quaternion multiplication. This group is generated by i and j, where

$$i^4 = 1, \qquad j^2 = i^2 \qquad \text{and} \qquad ji = i^3 j.$$

There are no finite strictly skew fields. This is the content of a famous theorem of Wedderburn, which we state without proof.

24.10 Theorem **(Wedderburn's Theorem)** Every finite division ring is a field.

Proof See Artin, Nesbitt, and Thrall [24] for proof of Wedderburn's theorem. ◆

EXERCISES 24

Computations

In Exercises 1 through 3, let $G = \{e, a, b\}$ be a cyclic group of order 3 with identity element e. Write the element in the group algebra $\mathbb{Z}_5 G$ in the form

$$re + sa + tb \qquad \text{for} \qquad r, s, t \in \mathbb{Z}_5.$$

1. $(2e + 3a + 0b) + (4e + 2a + 3b)$ **2.** $(2e + 3a + 0b)(4e + 2a + 3b)$ **3.** $(3e + 3a + 3b)^4$

In Exercises 4 through 7, write the element of $\mathbb{H}$ in the form $a_1 + a_2 i + a_3 j + a_4 k$ for $a_i \in \mathbb{R}$.

4. $(i + 3j)(4 + 2j - k)$ **5.** $i^2 j^3 k j i^5$

6. $(i + j)^{-1}$ **7.** $[(1 + 3i)(4j + 3k)]^{-1}$

8. Referring to the group S_3 given in Example 8.7, compute the product

$$(0\rho_0 + 1\rho_1 + 0\rho_2 + 0\mu_1 + 1\mu_2 + 1\mu_3)(1\rho_0 + 1\rho_1 + 0\rho_2 + 1\mu_1 + 0\mu_2 + 1\mu_3)$$

in the group algebra $\mathbb{Z}_2 S_3$.

9. Find the center of the group $\langle \mathbb{H}^*, \cdot \rangle$, where $\mathbb{H}^*$ is the set of nonzero quaternions.

Concepts

10. Find two subsets of $\mathbb{H}$ different from $\mathbb{C}$ and from each other, each of which is a field isomorphic to $\mathbb{C}$ under the induced addition and multiplication from $\mathbb{H}$.

11. Mark each of the following true or false.

_____ **a.** $M_n(F)$ has no divisors of 0 for any n and any field F.

_____ **b.** Every nonzero element of $M_2(\mathbb{Z}_2)$ is a unit.

_____ **c.** End(A) is always a ring with unity $\neq 0$ for every abelian group A.

_____ **d.** End(A) is never a ring with unity $\neq 0$ for any abelian group A.

_____ **e.** The subset Iso(A) of End(A), consisting of the isomorphisms of A onto A, forms a subring of End(A) for every abelian group A.

_____ **f.** $R\langle\mathbb{Z}, +\rangle$ is isomorphic to $\langle Z, +, \cdot\rangle$ for every commutative ring R with unity.

_____ **g.** The group ring RG of an abelian group G is a commutative ring for any commutative ring R with unity.

_____ **h.** The quaternions are a field.

_____ **i.** $\langle\mathbb{H}^*, \cdot\rangle$ is a group where $\mathbb{H}^*$ is the set of nonzero quaternions.

_____ **j.** No subring of $\mathbb{H}$ is a field.

12. Show each of the following by giving an example.

a. A polynomial of degree n with coefficients in a strictly skew field may have more than n zeros in the skew field.

b. A finite multiplicative subgroup of a strictly skew field need not be cyclic.

Theory

13. Let ϕ be the element of End($\langle\mathbb{Z} \times \mathbb{Z}, +\rangle$) given in Example 24.2. That example showed that ϕ is a right divisor of 0. Show that ϕ is also a left divisor of 0.

14. Show that $M_2(F)$ has at least six units for every field F. Exhibit these units. [Hint: F has at least two elements, 0 and 1.]

15. Show that End ($\langle\mathbb{Z}, +\rangle$) is naturally isomorphic to $\langle\mathbb{Z}, +, \cdot\rangle$ and that End($\langle\mathbb{Z}_n, +\rangle$) is naturally isomorphic to $\langle\mathbb{Z}_n, +, \cdot\rangle$.

16. Show that End($\langle\mathbb{Z}_2 \times \mathbb{Z}_2, +\rangle$) is not isomorphic to $\langle\mathbb{Z}_2 \times \mathbb{Z}_2, +, \cdot\rangle$.

17. Referring to Example 24.3, show that $YX - XY = 1$.

18. If $G = \{e\}$, the group of one element, show that RG is isomorphic to R for any ring R.

19. There exists a matrix $K \in M_2(\mathbb{C})$ such that $\phi : \mathbb{H} \to M_2(\mathbb{C})$ defined by

$$\phi(a + bi + cj + dk) = a\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} + b\begin{bmatrix}0 & 1\\ -1 & 0\end{bmatrix} + c\begin{bmatrix}0 & i\\ i & 0\end{bmatrix} + dK,$$

for all $a, b, c, d \in \mathbb{R}$, gives an isomorphism of $\mathbb{H}$ with $\phi[\mathbb{H}]$

a. Find the matrix K.

b. What 8 equations should you check to see that ϕ really is a homomorphism?

c. What other thing should you check to show that ϕ gives an isomorphism of $\mathbb{H}$ with $\phi[\mathbb{H}]$?

SECTION 25 †Ordered Rings and Fields

We are familiar with the inequality relation $<$ on the set $\mathbb{R}$ and on any subset of $\mathbb{R}$. (We remind you that relations were discussed in Section 0. See Definition 0.7.) We regard $<$ as providing an *ordering* of the real numbers. In this section, we study orderings of rings and fields. *We assume throughout this section that the rings under discussion have nonzero unity* 1.

In the real numbers, $a < b$ if and only if $b - a$ is positive, so the order relation $<$ on $\mathbb{R}$ is completely determined if we know which real numbers are positive. We use the idea of labeling certain elements as positive to define the notion of order in a ring.

† This section is not used in the remainder of the text.

25.1 Definition An **ordered ring** is a ring R together with a nonempty subset P of R satisfying these two properties.

Closure For all $a, b \in P$, both $a + b$ and ab are in P.

Trichotomy For each $a \in R$, one and only one of the following holds:

$$a \in P, \quad a = 0, \quad -a \in P.$$

Elements of P are called **"positive."** ■

It is easy to see that if R is an ordered ring with set P of positive elements and S is a subring of R, then $P \cap S$ satisfies the requirements for a set of positive elements in the ring S, and thus gives an ordering of S. (See Exercise 26.) This is the **induced ordering** from the given ordering of R.

We observe at once that for each of the rings $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$ the set of elements that we have always considered to be positive satisfies the conditions of closure and trichotomy. We will refer to the familiar ordering of these rings and the induced ordering on their subrings as the *natural ordering*. We now give an unfamiliar illustration.

25.2 Example Let R be an ordered ring with set P of positive elements. There are two natural ways to define an ordering of the polynomial ring $R[x]$. We describe two possible sets, P_{low} and P_{high}, of positive elements. A nonzero polynomial in $R[x]$ can be written in the form

$$f(x) = a_r x^r + a_{r+1} x^{r+1} + \cdots + a_n x^n$$

where $a_r \neq 0$ and $a_n \neq 0$, so that $a_r x^r$ and $a_n x^n$ are the nonzero terms of lowest and highest degree, respectively. Let P_{low} be the set of all such $f(x)$ for which $a_r \in P$, and let P_{high} be the set of all such $f(x)$ for which $a_n \in P$. The closure and trichotomy requirements that P_{low} and P_{high} must satisfy to give orderings of $R[x]$ follow at once from those same properties for P and the definition of addition and multiplication in $R[x]$. Illustrating in $\mathbb{Z}[x]$, with ordering given by P_{low}, the polynomial $f(x) = -2x + 3x^4$ would not be positive because -2 is not positive in $\mathbb{Z}$. With ordering given by P_{high}, this same polynomial would be positive because 3 is positive in $\mathbb{Z}$. ▲

Suppose now that P is the set of positive elements in an ordered ring R. Let a be any nonzero element of R. Then either a or $-a$ is in P, so by closure, $a^2 = (-a)^2$ is also in P. Thus all squares of nonzero elements of R are positive. In particular, $1 = 1^2$ is positive. By closure, we see that $1 + 1 + \cdots + 1$ for any finite number of summands is always in P, so it is never zero. Thus an ordered ring has characteristic zero.

Because squares of nonzero elements must be positive, we see that the natural ordering of $\mathbb{R}$ is the only possible ordering. The positive real numbers are precisely the squares of nonzero real numbers and the set could not be enlarged without destroying trichotomy. Because $1 + 1 + \cdots + 1$ must be positive, the only possible ordering of $\mathbb{Z}$ is the natural ordering also. All ordered rings have characteristic zero so we can, by identification (renaming), consider every ordered ring to contain $\mathbb{Z}$ as an ordered subring.

If a and b are nonzero elements of P then either $-a$ or a is in P and either $-b$ or b is in P. Consequently by closure, either ab or $-ab$ is in P. By trichotomy, ab cannot be zero so an ordered ring can have no zero divisors.

We summarize these observations in a theorem and corollary.

25.3 Theorem Let R be an ordered ring. All squares of nonzero elements of R are positive. R has characteristic 0, and there are no zero divisors.

25.4 Corollary We can consider $\mathbb{Z}$ to be embedded in any ordered ring R, and the induced ordering of $\mathbb{Z}$ from R is the natural ordering of $\mathbb{Z}$. The only possible ordering of $\mathbb{R}$ is the natural ordering.

Theorem 25.3 shows that the field $\mathbb{C}$ of complex numbers cannot be ordered, because both $1 = 1^2$ and $-1 = i^2$ are squares. It also shows that no finite ring can be ordered because the characteristic of an ordered ring is zero.

The theorem that follows defines a relation $<$ in an ordered ring, and gives properties of $<$. The definition of $<$ is motivated by the observation that, in the real numbers, $a < b$ if and only if $b - a$ is positive. The theorem also shows that ordering could have been defined in terms of a relation $<$ having the listed properties.

25.5 Theorem Let R be an ordered ring with set P of positive elements. Let $<$, read "is less than," be the relation on R defined by

$$a < b \text{ if and only if } (b - a) \in P \tag{1}$$

for $a, b \in R$. The relation $<$ has these properties for all $a, b, c \in R$.

Trichotomy One and only one of the following holds:

$$a < b, \quad a = b, \quad b < a.$$

Transitivity If $a < b$ and $b < c$, then $a < c$.

Isotonicity If $b < c$, then $a + b < a + c$.
If $b < c$ and $0 < a$, then $ab < ac$ and $ba < ca$.

Conversely, given a relation $<$ on a nonzero ring R satisfying these three conditions, the set $P = \{x \in R \mid 0 < x\}$ satisfies the two criteria for a set of positive elements in Definition 25.1, and the relation $<_P$ defined as in Condition (1) with this P is the given relation $<$.

Proof Let R be an ordered ring with set P of positive elements, and let $a < b$ mean $(b - a) \in P$. We prove the three properties for $<$.

Trichotomy Let $a, b \in R$. By the trichotomy property of P in Definition 25.1 applied to $b - a$, exactly one of

$$(b - a) \in P, \quad b - a = 0, \quad (a - b) \in P$$

holds. These translate in terms of $<$ to

$$a < b, \quad a = b, \quad b < a$$

respectively.

Transitivity Let $a < b$ and $b < c$. Then $(b - a) \in P$ and $(c - b) \in P$. By closure of P under addition, we have

$$(b - a) + (c - b) = (c - a) \in P$$

so $a < c$.

Isotonicity Let $b < c$, so $(c - b) \in P$. Then $(a + c) - (a + b) = (c - b) \in P$ so $a + b < a + c$. Also if $a > 0$, then by closure of P both $a(c - b) = ac - ab$ and $(c - b)a = ca - ba$ are in P, so $ab < ac$ and $ba < ca$.

We leave the "conversely" part of the theorem as an equally easy exercise. (See Exercise 27.) ◆

In view of Theorem 25.5, we will now feel free to use the $<$ notation in an ordered ring. The notations $>$, $\leq$, and $\geq$ are defined as usual in terms of $<$ and $=$. Namely,

$$b > a \text{ means } a < b, \quad a \leq b \text{ means either } a = b \text{ or } a < b,$$
$$a \geq b \text{ means either } b < a \text{ or } b = a.$$

25.6 Example Let R be an ordered ring. It is illustrative to think what the orderings of $R[x]$ given by P_{low} and P_{high} in Example 25.2 mean in terms of the relation $<$ of Theorem 25.5.

Taking P_{low}, we observe, for every $a > 0$ in R, that $a - x$ is positive so $x < a$. Also, $x = x - 0$ is positive, so $0 < x$. Thus $0 < x < a$ for every $a \in R$. We have $(x^i - x^j) \in P_{\text{low}}$ when $i < j$, so $x^j < x^i$ if $i < j$. Our monomials have the ordering

$$0 < \cdots x^6 < x^5 < x^4 < x^3 < x^2 < x < a$$

for any positive $a \in R$. Taking $R = \mathbb{R}$, we see that in this ordering of $\mathbb{R}[x]$ there are infinitely many positive elements that are less than any positive real number!

We leave a similar discussion of $<$ for the ordering of $R[x]$ given by P_{high} to Exercise 1. ▲

The preceding example is of interest because it exhibits an ordering that is *not Archimedian*. We give a definition explaining this terminology. Remember that we can consider $\mathbb{Z}$ to be a subring of every ordered ring.

25.7 Definition An ordering of a ring R with this property:

> For each given positive a and b in R, there exists a positive integer n such that $na > b$.

is an **Archimedian ordering.** ■

The natural ordering of $\mathbb{R}$ is Archimedian, but the ordering of $\mathbb{R}[x]$ given by P_{low} discussed in Example 25.6 is not Archimedian because for every positive integer n we have $(17 - nx) \in P_{\text{low}}$, so $nx < 17$ for all $n \in \mathbb{Z}^+$.

We give two examples describing types of ordered rings and fields that are of interest in more advanced work.

25.8 Example **(Formal Power Series Rings)** Let R be a ring. In Section 22 we defined a polynomial in $R[x]$ to be a formal sum $\sum_{i=0}^{\infty} a_i x^i$ where all but a finite number of the a_i are 0. If we do not require any of the a_i to be zero, we obtain a **formal power series** in x with coefficients in the ring R. (The adjective, *formal*, is customarily used because we are not dealing with convergence of series.) Exactly the same formulas are used to define the sum and product of these series as for polynomials in Section 22. Most of us had some

practice adding and multiplying series when we studied calculus. These series form a ring which we denote by $R[[x]]$, and which contains $R[x]$ as a subring.

If R is an ordered ring, we can extend the ordering to $R[[x]]$ exactly as we extended the ordering to $R[x]$ using the set P_{low} of positive elements. (We cannot use P_{high}. Why not?) The monomials have the same ordering that we displayed in Example 25.6. ▲

25.9 Example **(Formal Laurent Series Fields)** Continuing with the idea of Example 25.8, we let F be a field and consider formal series of the form $\sum_{i=N}^{\infty} a_i x^i$ where N may be any integer, positive, zero, or negative, and $a_i \in F$. (Equivalently, we could consider $\sum_{i=-\infty}^{\infty} a_i x^i$ where all but a finite number of the a_i are zero for negative values of i. In studying calculus for functions of a complex variable, one encounters series of this form called "*Laurent series*.") With the natural addition and multiplication of these series, we actually have a *field* which we denote by $F((x))$. The inverse of x is the series $x^{-1} + 0 + 0x + 0x^2 + \cdots$. Inverses of elements and quotients can be computed by series division. We compute three terms of $(x^{-1} - 1 + x - x^2 + x^3 + \cdots)/(x^3 + 2x^4 + 3x^5 + \cdots)$ in $\mathbb{R}((x))$ for illustration.

$$
\begin{array}{r|l}
 & x^{-4} - 3x^{-3} + 4x^{-2} + \cdots \\
\cline{2-2}
x^3 + 2x^4 + 3x^5 + \cdots & x^{-1} - 1 + x - x^2 + x^3 + \cdots \\
 & \underline{x^{-1} + 2 + 3x + \cdots} \\
 & \quad -3 - 2x + \cdots \\
 & \quad \underline{-3 - 6x - 9x^2 + \cdots} \\
 & \qquad 4x + \cdots
\end{array}
$$

If F is an ordered field, we can use the obvious analog of P_{low} in $R[[x]]$ to define an ordering of $F((x))$. In Exercise 2 we ask you to symbolically order the monomials $\cdots x^{-3}, x^{-2}, x^{-1}, x^0 = 1, x, x^2, x^3, \cdots$ as we did for $R[x]$ in Example 25.6. Note that $F((x))$ contains, as a subfield, a field of quotients of $F[x]$, and thus induces an ordering on this field of quotients. ▲

Let R be an ordered ring and let $\phi : R \to R'$ be a ring isomorphism. It is intuitively clear that by identification (renaming), the map ϕ can be used to carry over the ordering of R to provide an ordering of R'. We state as a theorem what would have to be proved for a skeptic, and leave the proof as Exercise 25.

25.10 Theorem Let R be an ordered ring with set P of positive elements and let $\phi : R \to R'$ be a ring isomorphism. The subset $P' = \phi[P]$ satisfies the requirements of Definition 25.1 for a set of positive elements of R'. Furthermore, in the ordering of R' given by P', we have $\phi(a) <' \phi(b)$ in R' if and only if $a < b$ in R.

We call the ordering of R' described in the preceding theorem the **"ordering induced by"** ϕ from the ordering of R.

25.11 Example Example 22.9 stated that the evaluation homomorphism $\phi_\pi : \mathbb{Q}[x] \to \mathbb{R}$ where

$$\phi(a_0 + a_1 x + \cdots + a_n x^n) = a_0 + a_1 \pi + \cdots + a_n \pi^n$$

is one to one. Thus it provides an isomorphism of $\mathbb{Q}[x]$ with $\phi[\mathbb{Q}[x]]$. We denote this

image ring by $\mathbb{Q}[\pi]$. If we provide $\mathbb{Q}[x]$ with the ordering using the set P_{low} of Examples 25.2 and 25.6, the ordering on $\mathbb{Q}[\pi]$ induced by ϕ_π is very different from that induced by the natural (and only) ordering of $\mathbb{R}$. In the P_{low} ordering, π is less than any element of $\mathbb{Q}$! ▲

An isomorphism of a ring R onto itself is called an **automorphism** of R. Theorem 25.10 can be used to exhibit different orderings of an ordered ring R if there exist automorphisms of R that do not carry the set P of positive elements onto itself. We give an example.

25.12 Example Exercise 11 of Section 18 shows that $\{m + n\sqrt{2} \mid m, n \in \mathbb{Z}\}$ is a ring. Let us denote this ring by $\mathbb{Z}[\sqrt{2}]$. This ring has a natural order induced from $\mathbb{R}$ in which $\sqrt{2}$ is positive. However, we claim that $\phi : \mathbb{Z}[\sqrt{2}] \to \mathbb{Z}[\sqrt{2}]$ defined by $\phi(m + n\sqrt{2}) = m - n\sqrt{2}$ is an automorphism. It is clearly one to one and onto $\mathbb{Z}[\sqrt{2}]$. We leave the verification of the homomorphism property to Exercise 17. Because $\phi(\sqrt{2}) = -\sqrt{2}$, we see the ordering induced by ϕ will be one where $-\sqrt{2}$ is positive! In the natural order on $\mathbb{Z}[\sqrt{2}]$, an element $m + n\sqrt{2}$ is positive if m and n are both positive, or if m is positive and $2n^2 < m^2$, or if n is positive and $m^2 < 2n^2$. In Exercise 3, we ask you to give the analogous descriptions for positive elements in the ordering of $\mathbb{Z}[\sqrt{2}]$ induced by ϕ. ▲

In view of Examples 25.11 and 25.12, which exhibit orderings on subrings of $\mathbb{R}$ that are not the induced orderings, we wonder whether $\mathbb{Q}$ can have an ordering other than the natural one. Our final theorem shows that this is not possible.

25.13 Theorem Let D be an ordered integral domain with P as set of positive elements, and let F be a field of quotients of D. The set

$$P' = \{x \in F \mid x = a/b \text{ for } a, b \in D \text{ and } ab \in P\}$$

is well-defined and gives an order on F that induces the given order on D. Furthermore, P' is the only subset of F with this property.

Proof To show that P' is well-defined, suppose that $x = a/b = a'/b'$ for $a, b, a', b' \in D$ and that $ab \in P$. We must show that $a'b' \in P$. From $a/b = a'/b'$ we obtain $ab' = a'b$. Multiplying by b, we have $(ab)b' = a'b^2$. Now $b^2 \in P$ and by assumption, $ab \in P$. Using trichotomy and the properties $a(-b) = (-a)b = -(ab)$ of a ring, we see that either a' and b' are both in P or both not in P. In either case, we have $a'b' \in P$.

We proceed to closure for P'. Let $x = a/b$ and $y = c/d$ be two elements of P', so $ab \in P$ and $cd \in P$. Now $x + y = (ad + bc)/bd$ and $(ad + bc)bd = (ab)d^2 + b^2(cd)$ is in P because squares are also in P and P is closed under addition and multiplication. Thus $(x + y) \in P'$. Also $xy = ac/bd$ is in P' because $acbd = (ab)(cd)$ is a product of two elements of P and thus in P.

For trichotomy, we need only observe that for $x = a/b$, the product ab satisfies just one of

$$ab \in P, \qquad ab = 0, \qquad ab \notin P$$

by trichotomy for P. For P', these translate into $x \in P'$, $x = 0$, and $x \notin P'$, respectively.

We have shown that P' does give an ordering of F. For $a \in D$, we see that $a = a/1$ is in P' if and only if $a1 = a$ is in P, so the given ordering on D is indeed the induced ordering from F by P'.

Finally, suppose that P'' is a set of positive elements of F satisfying the conditions of Definition 25.1 and such that $P'' \cap D = P$. Let $x = a/b \in P''$ where $a, b \in D$. Then $xb^2 = ab$ must be in P'', so $ab \in (P'' \cap D) = P$. Thus $x \in P'$ so $P'' \subseteq P'$. The law of trichotomy shows that we then must have $P' = P''$. Therefore P' gives the only ordering of F that maintains original order for elements of D. ◆

EXERCISES 25

Computations

1. Let R be an ordered ring. Describe the ordering of a positive element a of R and the monomials $x, x^2, x^3, \cdots, x^n, \cdots$ in $R[x]$ as we did in Example 25.6, but using the set P_{high} of Example 25.6 as set of positive elements of $R[x]$.
2. Let F be an ordered field and let $F((x))$ be the field of formal Laurent series with coefficients in F, discussed in Example 25.9. Describe the ordering of the monomials $\cdots x^{-3}, x^{-2}, x^{-1}, x^0 = 1, x, x^2, x^3, \cdots$ in the ordering of $F((x))$ described in that example.
3. Example 25.12 described an ordering of $\mathbb{Z}[\sqrt{2}] = \{m + n\sqrt{2} \mid m, n \in \mathbb{Z}\}$ in which $-\sqrt{2}$ is positive. Describe, in terms of m and n, all positive elements of $\mathbb{Z}[\sqrt{2}]$ in that ordering.

In Exercises 4 through 9, let $\mathbb{R}[x]$ have the ordering given by

i. P_{low} ii. P_{high}

as described in Example 25.2. In each case (i) and (ii), list the labels a, b, c, d, e of the given polynomials in an order corresponding to increasing order of the polynomials as described by the relation $<$ of Theorem 25.5.

4.	**a.** $-5 + 3x$	**b.** $5 - 3x$	**c.** $-x + 7x^2$	**d.** $x - 7x^2$	**e.** $2 + 4x^2$
5.	**a.** -1	**b.** $3x - 8x^3$	**c.** $-5x + 7x^2 - 11x^4$	**d.** $8x^2 + x^5$	**e.** $-3x^3 - 4x^5$
6.	**a.** $-3 + 5x^2$	**b.** $-2x + 5x^2 + x^3$	**c.** -5	**d.** $6x^3 + 8x^4$	**e.** $8x^4 - 5x^5$
7.	**a.** $-2x^2 + 5x^3$	**b.** $x^3 + 4x^4$	**c.** $2x - 3x^2$	**d.** $-3x - 4x^2$	**e.** $2x - 2x^2$
8.	**a.** $4x - 3x^2$	**b.** $4x + 2x^2$	**c.** $4x - 6x^3$	**d.** $5x - 6x^3$	**e.** $3x - 2x^2$
9.	**a.** $x - 3x^2 + 5x^3$	**b.** $2 - 3x^2 + 5x^3$	**c.** $x - 3x^2 + 4x^3$	**d.** $x + 3x^2 + 4x^4$	**e.** $x + 3x^2 - 4x^3$

In Exercises 10 through 13, let $\mathbb{Q}((x))$ have the ordering described in Example 25.9. List the labels a, b, c, d, e of the given elements in an order corresponding to increasing order of the elements as described by the relation $<$ of Theorem 25.5.

10.	**a.** $\dfrac{1}{x}$	**b.** $\dfrac{-5}{x^2}$	**c.** $\dfrac{2}{x}$	**d.** $\dfrac{-3}{x^2}$	**e.** $4x$
11.	**a.** $\dfrac{1}{1-x}$	**b.** $\dfrac{x^2}{1+x}$	**c.** $\dfrac{1}{x-x^2}$	**d.** $\dfrac{-x}{1+x^2}$	**e.** $\dfrac{3-2x}{x^3+4x}$
12.	**a.** $\dfrac{5-7x}{x^2+3x^3}$	**b.** $\dfrac{-2+4x}{4-3x}$	**c.** $\dfrac{7+2x}{4-3x}$	**d.** $\dfrac{9-3x^2}{2+6x}$	**e.** $\dfrac{3-5x}{-6+2x}$

13. **a.** $\dfrac{1-x}{1+x}$ **b.** $\dfrac{3-5x}{3+5x}$ **c.** $\dfrac{1}{4x+x^2}$ **d.** $\dfrac{1}{-3x+x^2}$ **e.** $\dfrac{4x+x^2}{1-x}$

Concepts

14. It can be shown that the smallest subfield of $\mathbb{R}$ containing $\sqrt[3]{2}$ is isomorphic to the smallest subfield of $\mathbb{C}$ containing $\sqrt[3]{2}(\frac{-1+i\sqrt{3}}{2})$. Explain why this shows that, although there is no ordering for $\mathbb{C}$, there may be an ordering of a subfield of $\mathbb{C}$ that contains some elements that are not real numbers.

15. Mark each of the following true or false.

_____ **a.** There is only one ordering possible for the ring $\mathbb{Z}$.

_____ **b.** The field $\mathbb{R}$ can be ordered in only one way.

_____ **c.** Any subfield of $\mathbb{R}$ can be ordered in only one way.

_____ **d.** The field $\mathbb{Q}$ can be ordered in only one way.

_____ **e.** If R is an ordered ring, then $R[x]$ can be ordered in a way that induces the given order on R.

_____ **f.** An ordering of a ring R is Archimedian if for each $a, b \in R$, there exists $n \in \mathbb{Z}^+$ such that $b < na$.

_____ **g.** An ordering of a ring R is Archimedian if for each $a, b \in R$ such that $0 < a$, there exists $n \in \mathbb{Z}^+$ such that $b < na$.

_____ **h.** If R is an ordered ring and $a \in R$, then $-a$ cannot be positive.

_____ **i.** If R is an ordered ring and $a \in R$, then either a or $-a$ is positive.

_____ **j.** Every ordered ring has an infinite number of elements.

16. Describe an ordering of the ring $\mathbb{Q}[\pi]$, discussed in Example 25.11, in which π is greater than any rational number.

Theory

17. Referring to Example 25.12, show that the map $\phi : \mathbb{Z}[\sqrt{2}] \to \mathbb{R}$ where $\phi(m + n\sqrt{2}) = m - n\sqrt{2}$ is a homomorphism.

In Exercises 18 through 24, let R be an ordered ring with set P of positive elements, and let $<$ be the relation on R defined in Theorem 25.5. Prove the given statement. (All the proofs have to be in terms of Definition 25.1 and Theorem 25.5. For example, you must not say, "We know that negative times positive is negative, so if $a < 0$ and $0 < b$ then $ab < 0$.")

18. If $a \in P$, then $0 < a$.

19. If $a, b \in P$ and $ac = bd$, then either $c = d = 0$ or $cd \in P$.

20. If $a < b$, then $-b < -a$.

21. If $a < 0$ and $0 < b$, then $ab < 0$.

22. If R is a field and a and b are positive, then a/b is positive.

23. If R is a field and $0 < a < 1$, then $1 < 1/a$.

24. If R is a field and $-1 < a < 0$, then $1/a < -1$.

25. Prove Theorem 25.10 of the text.

26. Show that if R is an ordered ring with set P of positive elements and S is a subring of R, then $P \cap S$ satisfies the requirements for a set of positive elements in the ring S, and thus gives an ordering of S.

27. Show that if $<$ is a relation on a ring R satisfying the properties of trichotomy, transitivity, and isotonicity stated in Theorem 25.5, then there exists a subset P of R satisfying the conditions for a set of positive elements

in Definition 25.1, and such that the relation $<_P$ defined by $a <_P b$ if and only if $(b - a) \in P$ is the same as the relation $<$.

28. Let R be an ordered integral domain. Show that if $a^{2n+1} = b^{2n+1}$ where $a, b \in R$ and n is a positive integer, then $a = b$.

29. Let R be an ordered ring and consider the ring $R[x, y]$ of polynomials in two variables with coefficients in R. Example 25.2 describes two ways in which we can order $R[x]$, and for each of these, we can continue on and order $(R[x])[y]$ in the analogous two ways, giving four ways of arriving at an ordering of $R[x, y]$. There are another four ways of arriving at an ordering of $R[x, y]$ if we first order $R[y]$ and then $(R[y])[x]$. Show that all eight of these orderings of $R[x, y]$ are different. [*Hint:* You might start by considering whether $x < y$ or $y < x$ in each of these orderings, and continue in this fashion.]